Project:

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2025-05-20

Project Setup

```
# Load libraries
library(SNPRelate)

## Loading required package: gdsfmt

## SNPRelate -- supported by Streaming SIMD Extensions 2 (SSE2)

library(GENESIS)
library(GWASTools)

## Loading required package: Biobase

## Loading required package: BiocGenerics

##
## Attaching package: 'BiocGenerics'

## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs

## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
##     match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
##     Position, rank, rbind, Reduce, rownames, sapply, saveRDS, setdiff,
##     table, tapply, union, unique, unsplit, which.max, which.min

## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname")'.

library(qqman)

## Warning: package 'qqman' was built under R version 4.4.3

##

## For example usage please run: vignette('qqman')
```

```
##

## Citation appreciated but not required:

## Turner, (2018). qqman: an R package for visualizing GWAS results using Q-Q and manhattan plots. Journal of Open Source Software, 3(25), 731, https://doi.org/10.21105/joss.00731.

##

library(GeneNet)

## Warning: package 'GeneNet' was built under R version 4.4.3

## Loading required package: corpcor

## Loading required package: longitudinal

## Loading required package: fdrtool

library(gplots)

## Warning: package 'gplots' was built under R version 4.4.3

##
## Attaching package: 'gplots'

## The following object is masked from 'package:stats':
##
##     lowess

library(readxl)

## Warning: package 'readxl' was built under R version 4.4.3

library(writexl)

## Warning: package 'writexl' was built under R version 4.4.3

library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following object is masked from 'package:Biobase':
##
##     combine

## The following objects are masked from 'package:BiocGenerics':
##
##     combine, intersect, setdiff, union
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

library(tidyr)
```

Task 1: Compute Kinship using SNPRelate and GENESIS

```
# Convert PLINK files to GDS format
snpgdsPED2GDS("Qatari156_filtered_pruned.ped",
               "Qatari156_filtered_pruned.map",
               "Qatari156_filtered_pruned.gds",
              family=TRUE)

## PLINK PED/MAP to GDS Format:
## Import 67735 variants from 'Qatari156_filtered_pruned.map'
## Chromosome:
##    1    2    3    4    5    6    7    8    9   10   11   12   13   14   15
16
## 5224 5144 4369 4124 4254 3976 3517 3346 3043 3553 3131 3412 2577 2303 2204
2316
##   17   18   19   20   21   22   23
## 1990 2214 1283 1895 1074 1129 1657
## Reading 'Qatari156_filtered_pruned.ped'
## Output: 'Qatari156_filtered_pruned.gds'
## Import 156 samples
## Transpose the genotypic matrix ...
## Done.
## Optimize the access efficiency ...
## Clean up the fragments of GDS file:
##     open the file 'Qatari156_filtered_pruned.gds' (5.7M)
##     # of fragments: 52
##     save to 'Qatari156_filtered_pruned.gds.tmp'
##     rename 'Qatari156_filtered_pruned.gds.tmp' (3.1M, reduced: 2.5M)
##     # of fragments: 26

# Open the GDS file
genofile <- snpgdsOpen("Qatari156_filtered_pruned.gds")

# Calculate IBD coefficients
ibd <- snpgdsIBDKING(genofile, num.thread=2)

## IBD analysis (KING method of moment) on genotypes:
## Excluding 1,657 SNPs on non-autosomes
## Excluding 0 SNP (monomorphic: TRUE, MAF: NaN, missing rate: NaN)
##     # of samples: 156
##     # of SNPs: 66,078
```

```
##     using 2 threads
## No family is specified, and all individuals are treated as singletons.
## Relationship inference in the presence of population stratification.
## KING IBD:    the sum of all selected genotypes (0,1,2) = 16145834
## CPU capabilities: Double-Precision SSE2
## Tue May 20 03:13:40 2025    (internal increment: 65536)
## [..................................................]  0%, ETC: ---
[==================================================] 100%, completed, 0s
## Tue May 20 03:13:40 2025    Done.

# Close the file
snpgdsClose(genofile)

# Extract kinship matrix
kinship <- ibd$kinship
rownames(kinship) <- ibd$sample.id
colnames(kinship) <- ibd$sample.id

saveRDS(ibd, file = "ibd.rds")
```

1b. Report individuals with kinship > 0.1

```
# Find pairs with kinship > 0.1
high_kinship_pairs <- which(kinship > 0.1, arr.ind = TRUE)
high_kinship_ind <- unique(rownames(high_kinship_pairs))

# Count and report
num_high_kinship <- length(high_kinship_ind)
cat("Number of individuals with kinship > 0.1:", num_high_kinship, "\n")

## Number of individuals with kinship > 0.1: 156

# Open GDS file

genofile <- snpgdsOpen("Qatari156_filtered_pruned.gds")

# Compute IBD-based kinship using MLE
ibd_mle <- snpgdsIBDMLE(genofile, kinship = TRUE, num.thread = 12)

## Identity-By-Descent analysis (MLE) on genotypes:
## Excluding 1,657 SNPs on non-autosomes
## Excluding 0 SNP (monomorphic: TRUE, MAF: NaN, missing rate: NaN)
##     # of samples: 156
##     # of SNPs: 66,078
##     using 12 threads
## MLE IBD:    the sum of all selected genotypes (0,1,2) = 16145834
## MLE IBD: Tue May 20 03:13:40 2025    0%
## MLE IBD: Tue May 20 03:14:10 2025    2%
## MLE IBD: Tue May 20 03:33:59 2025    100%
```

```r
# Extract kinship matrix
kinship_mat <- snpgdsIBDSelection(ibd_mle)

# Convert to matrix
kinship_df <- kinship_mat$kinship
rownames(kinship_df) <- colnames(kinship_df) <- kinship_mat$sample.id

high_kin <- kinship_mat[kinship_mat$kinship > 0.1, ]
unique_individuals <- unique(c(high_kin$ID1, high_kin$ID2))
length(unique_individuals)

## [1] 9

snpgdsClose(genofile)
```

Task 2: Compute mQTLs with Mixed Models

```r
# ----------------------------
# Step 1: Load PLINK PCA Results
# ----------------------------
eigenvectors <- read.table("Qatari156_PCA.eigenvec", header = FALSE)
colnames(eigenvectors) <- c("FID", "IID", paste0("PC", 1:(ncol(eigenvectors)-
2)))
eigenvalues <- read.table("Qatari156_PCA.eigenval", header = FALSE)

# ----------------------------
# Step 2: Load Metabolite Data
# ----------------------------
metabolite_data <- read.csv("qatari_metabolites_2025.csv")
rownames(metabolite_data) <- metabolite_data$Sample
pcs <- eigenvectors[, 2:5]
metabolite_data <- merge(metabolite_data, pcs, by.x = "Sample", by.y = "IID")
metabolite_data$scanID <- metabolite_data$Sample
scanAnnot <- ScanAnnotationDataFrame(metabolite_data)

sample_ids <- metabolite_data$scanID
kinship_matrix <- matrix(0, nrow = length(sample_ids), ncol =
length(sample_ids),
                         dimnames = list(sample_ids, sample_ids))
metabolite_names <- colnames(metabolite_data)[2:22]

for (i in 1:nrow(kinship_mat)) {
  id1 <- kinship_mat$ID1[i]
  id2 <- kinship_mat$ID2[i]
  kinship_matrix[id1, id2] <- kinship_mat$kinship[i]
  kinship_matrix[id2, id1] <- kinship_mat$kinship[i]
}

diag(kinship_matrix) <- 0.5
```

```
corrected_matrix <- data.frame(scanID =rownames(metabolite_data["scanID"]))
corrected_matrix$scanID <- metabolite_data$Sample
```

2a-c. mQTL analysis with mixed models

```
# -----------------------------
# Load Required Libraries
# -----------------------------
library(GENESIS)
library(dplyr)
library(ggplot2)
genoReader <- GdsGenotypeReader("Qatari156_filtered_pruned.gds")
# Create GenotypeData object
genoData <- GenotypeData(genoReader)
genoIter <- GenotypeBlockIterator(genoData, snpBlock = 100)
# -----------------------------
# Step 3: Define Variables
# -----------------------------
heritabilities <- data.frame(Metabolite = character(), Heritability =
numeric(), Lower = numeric(), Upper = numeric(), stringsAsFactors = FALSE)
inflation_factors <- data.frame(Metabolite = character(), Lambda = numeric(),
stringsAsFactors = FALSE)
all_assoc_results <- list()
significant_assoc_results <- list()


for (metabolite in metabolite_names) {
  phenotype <- metabolite_data[[metabolite]]

  sample_metabolite <- data.frame(
    scanID = metabolite_data$scanID,
    phenotype = phenotype,
    PC1 = metabolite_data$PC1,
    PC2 = metabolite_data$PC2,
    PC3 = metabolite_data$PC3,
    row.names = sample_ids
  )


  scanAnnot <- ScanAnnotationDataFrame(sample_metabolite)
  kin <- kinship_matrix[sample_metabolite$scanID, sample_metabolite$scanID]
  genoIter <- GenotypeBlockIterator(genoData, snpBlock = 100)


  corrected_metabolite <- fitNullModel(
    x = scanAnnot,
    outcome = "phenotype",
    covars = c("PC1", "PC2", "PC3"),
    cov.mat = kin,
```

```r
    family = "gaussian"
  )

  corrected_matrix[[metabolite]] <- corrected_metabolite$fit$resid.marginal

  # Estimate heritability
  heritability <- varCompCI(corrected_metabolite)
  h <- heritability[1, "Proportion"]
  h_lower <- heritability[1, "Lower 95"]
  h_upper <- heritability[1, "Upper 95"]


  heritabilities <- rbind(heritabilities, data.frame(
    Metabolite = metabolite,
    Heritability = h,
    Lower = h_lower,
    Upper = h_upper
  ))

  # Run association test
  assoc <- assocTestSingle(genoIter, null.model = corrected_metabolite)

  # Calculate inflation factor
  chisq <- qchisq(1 - assoc$Score.pval, df = 1)
  lambda <- median(chisq, na.rm = TRUE) / qchisq(0.5, df = 1)
  inflation_factors <- rbind(inflation_factors, data.frame(Metabolite =
metabolite, Lambda = lambda))

  # Add all associations
  assoc$Metabolite <- metabolite
  all_assoc_results[[metabolite]] <- assoc

  # Filter significant SNP-metabolite associations
  sig_assoc <- assoc[assoc$Score.pval < 1e-4, ]
  if (nrow(sig_assoc) > 0) {
    sig_assoc$Metabolite <- metabolite
    significant_assoc_results[[metabolite]] <- sig_assoc
  }

}
```

```
## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  140.150908  140.150908 -646.874628    1.350861
## [1]   21.354672  210.457918 -646.557174    1.269528
## [1]    7.279477  218.697709 -646.527236    1.261173
## [1]    0.328382  222.764840 -646.512966    1.257109
```

```
## [1]    0.1125786  222.8910805 -646.5125283    1.2569830
## [1]    0.000000  222.954187 -646.512300    1.256933
## [1]    0.000000  268.528849 -646.512300    1.043607
## [1]    0.000000  279.749204 -646.512300    1.001749
## [1]    0.000000  280.237632 -646.512300    1.000003
##              Proportion Lower 95 Upper 95
## V_A                   0       NA       NA
## V_resid.var           1        1        1

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  142.31419  142.31419 -648.93409    1.36687
## [1]  417.72541   64.47409 -648.82217    1.07896
## [1]  504.339488   32.141102 -648.808332    1.040985
## [1]  553.922749   13.520060 -648.804425    1.020921
## [1]  580.055431    3.735468 -648.803434    1.010575
## [1]  586.633755    1.291592 -648.803295    1.007939
## [1]  589.0900642    0.3820982 -648.8032535    1.0069476
## [1]  589.6259740    0.1839682 -648.8032453    1.0067305
## [1]  589.88538667    0.08809481 -648.80324142    1.00662534
## [1]  590.01302450    0.04093073 -648.80323954    1.00657358
## [1]  590.07633    0.00000 -648.80324    1.00661
## [1]  593.950907    0.000000 -648.803238    1.000043
##              Proportion Lower 95 Upper 95
## V_A                   1        1        1
## V_resid.var           0       NA       NA

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  138.26954  138.26954 -646.07942    1.35499
## [1]  411.716609   30.907443 -645.748661    1.200057
## [1]  475.975517    4.360440 -645.692907    1.175653
## [1]  483.557490    1.214955 -645.686793    1.172916
## [1]  485.4382023    0.4344023 -645.6852918    1.1722413
## [1]  486.37670238    0.04485858 -645.68454466    1.17190501
## [1]  486.435301    0.000000 -645.684459    1.171985
## [1]  557.818278    0.000000 -645.684459    1.022009
## [1]  569.830737    0.000000 -645.684459    1.000464
## [1]  570.0950    0.0000 -645.6845    1.0000
##              Proportion Lower 95 Upper 95
## V_A                   1        1        1
## V_resid.var           0       NA       NA

## Using 18 CPU cores
```

```
## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  160.421757  160.421757 -655.088129    1.314857
## [1]  389.875348   61.635530 -654.705739    1.241815
## [1]  499.680186   13.908812 -654.555711    1.212554
## [1]  525.881455    2.495458 -654.522693    1.205961
## [1]  529.112725    1.087391 -654.518692    1.205158
## [1]  530.7255707    0.3845445 -654.5167009    1.2047572
## [1]  531.53129438    0.03341862 -654.51570773    1.20455725
## [1]  531.581630    0.000000 -654.515613    1.204598
## [1]  621.869435    0.000000 -654.515613    1.029705
## [1]  639.809230    0.000000 -654.515613    1.000833
## [1]  640.341690    0.000000 -654.515613    1.000001
##             Proportion Lower 95 Upper 95
## V_A                  1        1        1
## V_resid.var          0       NA       NA

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  172.144976  172.144976 -659.273384    1.294684
## [1]   35.702989  267.246050 -659.178850    1.163019
## [1]    6.584969  286.380269 -659.166335    1.143013
## [1]    3.211298  288.586812 -659.165016    1.140783
## [1]    1.540975  289.678800 -659.164372    1.139685
## [1]    0.7099403  290.2219800 -659.1640543    1.1391392
## [1]    0.2954546  290.4928674 -659.1638964    1.1388675
## [1]    0.08846968  290.62813550 -659.16381766    1.13873190
## [1]    0.03675566  290.66193060 -659.16379801    1.13869803
##              Proportion  Lower 95 Upper 95
## V_A         0.000126439 -5.033504 5.033756
## V_resid.var 0.999873561 -4.033756 6.033504

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  121.245311  121.245311 -634.579372    1.328256
## [1]  233.338879  111.172919 -634.564210    1.065211
## [1]  319.532549   82.985884 -634.557998    1.003869
## [1]  349.929588   69.133397 -634.557133    1.000032
## [1]  358.335712   65.058841 -634.557062    1.000001
## [1]  360.70381   63.90936 -634.55706    1.00000
## [1]  361.37285   63.58455 -634.55706    1.00000
## [1]  361.56199   63.49272 -634.55706    1.00000
```

```
## [1]  361.61548   63.46675 -634.55706    1.00000
##              Proportion  Lower 95 Upper 95
## V_A           0.8506954 -1.210683 2.912073
## V_resid.var  0.1493046 -1.912073 2.210683

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  156.52596  156.52596 -655.64424    1.35748
## [1]   48.23097  216.13551 -655.41039    1.31459
## [1]   19.787112  231.612743 -655.354057    1.304855
## [1]    5.438761  239.401757 -655.326345    1.300114
## [1]    1.837479  241.354535 -655.319461    1.298943
## [1]    0.03515157  242.33157265 -655.31602637    1.29835984
##                Proportion  Lower 95 Upper 95
## V_A         0.0001450346 -5.631103 5.631393
## V_resid.var 0.9998549654 -4.631393 6.631103

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  123.996592  123.996592 -639.338760    1.382719
## [1]   60.871044  157.068136 -639.003531    1.359475
## [1]   28.060549  174.188048 -638.839321    1.348401
## [1]   11.386016  182.873897 -638.758257    1.342995
## [1]    2.987092  187.245608 -638.718008    1.340325
## [1]    0.8800161  188.3419618 -638.7079711    1.3396611
## [1]    0.3527979  188.6162597 -638.7054634    1.3394953
## [1]    0.08913293  188.75343478 -638.70420979    1.33941246
## [1]    0.00000  188.78773 -638.70379    1.33947
## [1]    0.000000  236.633363 -638.703786    1.068639
## [1]    0.000000  251.832293 -638.703786    1.004143
## [1]    0.000000  252.871221 -638.703786    1.000017
##             Proportion Lower 95 Upper 95
## V_A                  0       NA       NA
## V_resid.var          1        1        1

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  161.385244  161.385244 -656.142070    1.325259
## [1]   38.484860  227.189174 -655.739495    1.286463
## [1]    7.468218  243.720938 -655.648218    1.277610
```

```
## [1]    3.598531  245.781954 -655.637095    1.276527
## [1]    1.664369  246.812015 -655.631558    1.275988
## [1]    0.6974669  247.3269292 -655.6287948    1.2757181
## [1]    0.2140616  247.5843569 -655.6274148    1.2755835
## [1]    0.09321604  247.64871016 -655.62706997    1.27554983
## [1]    0.032794  247.680886 -655.626898    1.275533
##               Proportion  Lower 95 Upper 95
## V_A         0.0001323867 -4.660251 4.660515
## V_resid.var 0.9998676133 -3.660515 5.660251

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  119.222004  119.222004 -634.668187    1.352377
## [1]   72.31691  188.16110 -634.65008    1.07306
## [1]   34.572058  221.749578 -634.647680    1.004712
## [1]   33.875045  223.208366 -634.647680    1.000022
## [1]   33.97702  223.16392 -634.64768    1.00000
##             Proportion  Lower 95 Upper 95
## V_A          0.1321338 -4.621856 4.886124
## V_resid.var  0.8678662 -3.886124 5.621856

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  126.182791  126.182791 -639.710426    1.365424
## [1]   16.109325  191.722220 -639.503496    1.280448
## [1]    2.155260  199.834267 -639.481554    1.271753
## [1]    0.4183262  200.8420620 -639.4788802    1.2706921
## [1]    0.2013401  200.9679317 -639.4785470    1.2705599
## [1]    0.09285534  201.03085991 -639.47838052    1.27049387
## [1]    0.03861503  201.06232238 -639.47829728    1.27046085
## [1]    0.000000  201.078053 -639.478238    1.270479
## [1]    0.000000  243.886684 -639.478238    1.047476
## [1]    0.000000  254.940734 -639.478238    1.002059
## [1]    0.000000  255.464462 -639.478238    1.000004
##             Proportion Lower 95 Upper 95
## V_A                  0       NA       NA
## V_resid.var          1        1        1

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS
```

```
## [1]  150.583257  150.583257 -652.854433    1.360195
## [1]   46.569163  230.637049 -652.796187    1.201497
## [1]   18.016408  249.827625 -652.784980    1.176845
## [1]    4.077068  259.036820 -652.779997    1.165899
## [1]    0.6481105  261.2855537 -652.7788176    1.1633132
## [1]    0.2213884  261.5649227 -652.7786720    1.1629943
## [1]    0.000000  261.704499 -652.778597    1.162853
## [1]    0.000000  298.355137 -652.778597    1.020005
## [1]    0.000000  304.206731 -652.778597    1.000385
## [1]    0.0000  304.3237 -652.7786    1.0000
##             Proportion Lower 95 Upper 95
## V_A                  0       NA       NA
## V_resid.var          1        1        1

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  151.852679  151.852679 -651.778154    1.329857
## [1]   12.783350  275.359797 -651.735319    1.065982
## [1]    0.5352815  282.4135798 -651.7329250    1.0618756
## [1]    0.1585763  282.6292272 -651.7328528    1.0617547
## [1]    0.06444825  282.68310193 -651.73283481    1.06172456
## [1]    0.000000  282.710035 -651.732822    1.061741
## [1]    0.000000  299.149902 -651.732822    1.003393
## [1]    0.000000  300.161489 -651.732822    1.000011
##             Proportion Lower 95 Upper 95
## V_A                  0       NA       NA
## V_resid.var          1        1        1

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  126.628339  126.628339 -638.435559    1.337986
## [1]   50.251834  169.409202 -638.303894    1.297236
## [1]    8.702666  192.262902 -638.237318    1.278975
## [1]    3.321966  195.200179 -638.228912    1.276800
## [1]    0.620575  196.673553 -638.224709    1.275720
## [1]    0.2822175  196.8580172 -638.2241838    1.2755852
## [1]    0.1129962  196.9502678 -638.2239210    1.2755179
## [1]    0.02837488  196.99639756 -638.22378959    1.27548428
##               Proportion  Lower 95 Upper 95
## V_A         0.0001440168 -6.646524 6.646812
## V_resid.var 0.9998559832 -5.646812 7.646524

## Using 18 CPU cores
```

```
## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  122.896207  122.896207 -636.577454    1.345321
## [1]  273.275868   60.803516 -636.348322    1.264913
## [1]  347.745361   29.441973 -636.251661    1.232802
## [1]  383.793297   14.176274 -636.208468    1.218457
## [1]  401.431382    6.691432 -636.188159    1.211676
## [1]  410.14523    2.99038 -636.17832    1.20838
## [1]  414.474912    1.150673 -636.173483    1.206755
## [1]  416.632832    0.233580 -636.171083    1.205948
## [1]  417.171440    0.000000 -636.170473    1.205774
## [1]  488.364830    0.000000 -636.170473    1.029998
## [1]  502.587940    0.000000 -636.170473    1.000849
## [1]  503.014237    0.000000 -636.170473    1.000001
##             Proportion Lower 95 Upper 95
## V_A                  1        1        1
## V_resid.var          0       NA       NA

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  190.736956  190.736956 -669.908268    1.343988
## [1]  593.982205   66.385958 -669.797092    1.071266
## [1]  683.872814   28.574067 -669.779845    1.053801
## [1]  722.47785   12.12013 -669.77339    1.04713
## [1]  740.311629    4.480398 -669.770599    1.044195
## [1]  748.8795236    0.8017044 -669.7693015    1.0428158
## [1]  749.9292910    0.3504937 -669.7691444    1.0426486
## [1]  750.4528689    0.1254204 -669.7690662    1.0425653
## [1]  750.714332    0.000000 -669.769023    1.042561
## [1]  781.361120    0.000000 -669.769023    1.001669
## [1]  782.663304    0.000000 -669.769023    1.000003
##             Proportion Lower 95 Upper 95
## V_A                  1        1        1
## V_resid.var          0       NA       NA

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  155.733467  155.733467 -654.507323    1.344129
## [1]  355.166860   72.722910 -654.282169    1.263951
## [1]  462.90728   26.43672 -654.17315    1.23197
## [1]  517.572634    2.699657 -654.120544    1.217684
## [1]  520.996667    1.206249 -654.117301    1.216834
```

```
## [1]  522.7087181    0.4593385 -654.1156818    1.2164108
## [1]  523.5647486    0.0858339 -654.1148729    1.2161995
## [1]  523.67175257    0.03914279 -654.11477180    1.21617309
## [1]  523.725255    0.000000 -654.114687    1.216235
## [1]  616.838629    0.000000 -654.114687    1.032641
## [1]  636.3366    0.0000 -654.1147    1.0010
## [1]  636.972367    0.000000 -654.114687    1.000001
##             Proportion Lower 95 Upper 95
## V_A                  1        1        1
## V_resid.var          0       NA       NA

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  157.322856  157.322856 -653.464436    1.312417
## [1]  506.167423   13.312896 -652.972372    1.175852
## [1]  527.184477    4.255995 -652.950370    1.170391
## [1]  532.384927    2.012384 -652.945035    1.169062
## [1]  534.9782567    0.8932444 -652.9423906    1.1684022
## [1]  536.2731859    0.3343478 -652.9410742    1.1680734
## [1]  536.92021510    0.05506854 -652.94041747    1.16790933
## [1]  537.001066    0.000000 -652.940288    1.167979
## [1]  614.232527    0.000000 -652.940288    1.021121
## [1]  626.937420    0.000000 -652.940288    1.000428
## [1]  627.2056    0.0000 -652.9403    1.0000
##             Proportion Lower 95 Upper 95
## V_A                  1        1        1
## V_resid.var          0       NA       NA

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  136.049954  136.049954 -644.498831    1.348752
## [1]  140.972187  185.950193 -644.497171    1.071668
## [1]  122.330293  212.050481 -644.496435    1.004505
## [1]  110.592441  218.985607 -644.496320    1.000022
## [1]  106.8428  220.8184 -644.4963    1.0000
## [1]  105.7302  221.3605 -644.4963    1.0000
## [1]  105.4003  221.5212 -644.4963    1.0000
## [1]  105.3025  221.5689 -644.4963    1.0000
## [1]  105.2735  221.5830 -644.4963    1.0000
##             Proportion  Lower 95 Upper 95
## V_A          0.3220787 -4.319544 4.963702
## V_resid.var  0.6779213 -3.963702 5.319544

## Using 18 CPU cores
```

```
## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## [1]  142.960225  142.960225 -648.652339    1.355658
## [1]    9.028350  221.545197 -648.476636    1.272749
## [1]    4.438819  224.145937 -648.471299    1.270623
## [1]    2.141191  225.446944 -648.468641    1.269567
## [1]    0.9916964  226.0975906 -648.4673143    1.2690404
## [1]    0.4167828  226.4229477 -648.4666517    1.2687777
## [1]    0.1292849  226.5856344 -648.4663206    1.2686465
## [1]    0.0574054  226.6263071 -648.4662378    1.2686137
## [1]    0.000000  226.646644 -648.466172    1.268655
## [1]    0.000000  274.642226 -648.466172    1.046949
## [1]    0.000000  286.958271 -648.466172    1.002015
## [1]    0.000000  287.535339 -648.466172    1.000004
##             Proportion Lower 95 Upper 95
## V_A                  0       NA       NA
## V_resid.var          1        1        1

## Using 18 CPU cores

## Computing Variance Component Estimates...

## Sigma^2_A     log-lik     RSS

## Warning in log(2 * pi * RSS): NaNs produced
## Warning in log(2 * pi * RSS): NaNs produced

## [1]  3.225807e-03  3.225807e-03           NaN -1.882581e-16

## Warning in log(2 * pi * RSS): NaNs produced
## Warning in log(2 * pi * RSS): NaNs produced

## [1]  1.518910e-03  1.593092e-03           NaN -4.657032e-18

## Warning in log(2 * pi * RSS): NaNs produced
## Warning in log(2 * pi * RSS): NaNs produced

## [1]  3.402493e-03  2.878688e-04           NaN -3.034174e-16

## Warning in log(2 * pi * RSS): NaNs produced
## Warning in log(2 * pi * RSS): NaNs produced

## [1]  1.121408e-03  4.958018e-04           NaN -7.244206e-16
## [1] 1.050278e-04 4.177595e-04 2.942229e+03 1.857267e-15
## [1] 8.629083e-05 2.088536e-04 2.929746e+03 4.090853e-15
## [1] 3.756133e-05 1.791334e-04 2.875172e+03 1.065997e-14
## [1] 1.002140e-05 9.429596e-05 2.929409e+03 1.039545e-14
## [1] 0.000000e+00 8.114448e-06 3.847182e+03 7.236000e-19

## Warning in log(2 * pi * RSS): NaNs produced
## Warning in log(2 * pi * RSS): NaNs produced
```

```
## [1]  0.000000e+00  3.115906e-06           NaN -1.113175e-16

## Warning in log(2 * pi * RSS): NaNs produced
## Warning in log(2 * pi * RSS): NaNs produced

## [1]  0.00000e+00  7.94361e-07          NaN -1.70645e-12

## Warning in sqrt(diag(betaCov)): NaNs produced

##              Proportion Lower 95 Upper 95
## V_A                   0       NA       NA
## V_resid.var           1        1        1

## Using 18 CPU cores

# ------------------------------
# Step 5: Save Results
# ------------------------------
# Combine significant associations
library(openxlsx)

## Warning: package 'openxlsx' was built under R version 4.4.3

## Registered S3 method overwritten by 'openxlsx':
##   method               from
##   as.character.formula formula.tools

final_sig_assoc <- do.call(rbind,significant_assoc_results)
final_assoc <- do.call(rbind, all_assoc_results)
# Save each result to its own Excel file
write.xlsx(final_assoc, file = "mQTLs.xlsx", rowNames = FALSE)
write.xlsx(final_sig_assoc, file = "Significant_mQTLs.xlsx", rowNames =
FALSE)
write.xlsx(heritabilities, file = "Heritability.xlsx", rowNames = FALSE)
write.xlsx(inflation_factors, file = "Inflation_Factors.xlsx", rowNames =
FALSE)

library(openxlsx)
write.xlsx(corrected_matrix, file = "corrected_metabloite.xlsx", rowNames =
FALSE)
```

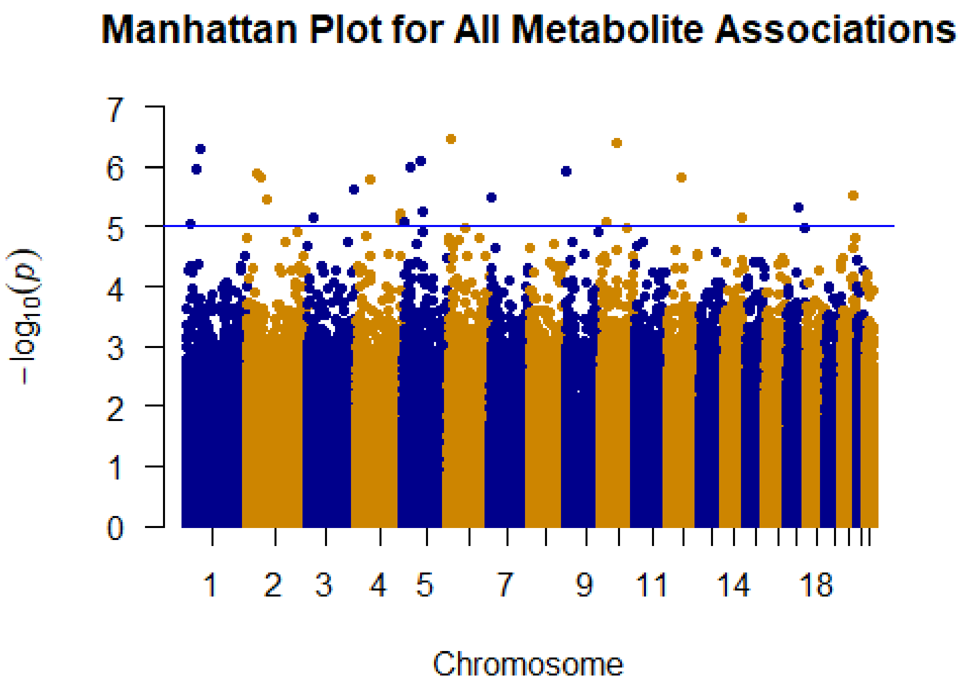
Task 3: Inflation Factor Calculation

```
# Calculate lambda (inflation factor) for each metabolite
lambda_values <- sapply(names(all_assoc_results), function(metab) {
    df <- all_assoc_results[[metab]]
    pvals <- df$Score.pval
    lambda <- median(qchisq(1 - pvals, df = 1)) / qchisq(0.5, 1)
    return(lambda)
})
# Calculate average lambda
avg_lambda <- mean(lambda_values)
```

```
cat("Average inflation factor:", avg_lambda, "\n")

## Average inflation factor: 1.00592
```

Task 4: Manhattan Plot

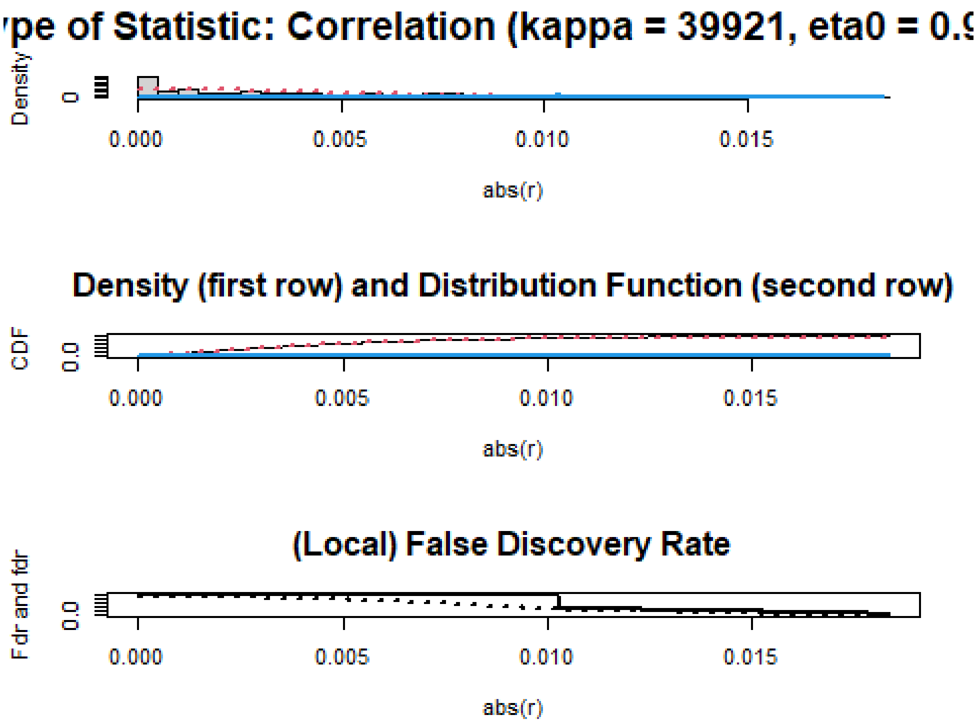
```
# Read the map file
snp_map <- read.table("Qatari156_filtered_pruned.map", header = FALSE,
stringsAsFactors = FALSE)

# Assign proper column names
colnames(snp_map) <- c("chr", "SNP", "cm", "pos")

# Make sure chr and pos are the correct types
snp_map$chr <- as.character(snp_map$chr)
snp_map$pos <- as.numeric(snp_map$pos)

for (metab in names(all_assoc_results)) {
  all_assoc_results[[metab]] <- merge(all_assoc_results[[metab]], snp_map,
                                      by = c("chr", "pos"),
                                      all.x = TRUE)
}

library(qqman)

# Combine all metabolite results into one data frame
all_assoc <- do.call(rbind, all_assoc_results)

# Rename for qqman
colnames(all_assoc)[colnames(all_assoc) == "chr"] <- "CHR"
colnames(all_assoc)[colnames(all_assoc) == "pos"] <- "BP"
colnames(all_assoc)[colnames(all_assoc) == "Score.pval"] <- "P"

# Ensure correct types
all_assoc$CHR <- as.numeric(all_assoc$CHR)

## Warning: NAs introduced by coercion

all_assoc$BP <- as.numeric(all_assoc$BP)

# Filter only needed columns
manhattan_df <- na.omit(all_assoc[, c("CHR", "BP", "SNP", "P")])

manhattan(manhattan_df,
          col = c("blue4", "orange3"),
          genomewideline = -log10(5e-8),
          suggestiveline = -log10(1e-5),
          main = "Manhattan Plot for All Metabolite Associations")
```



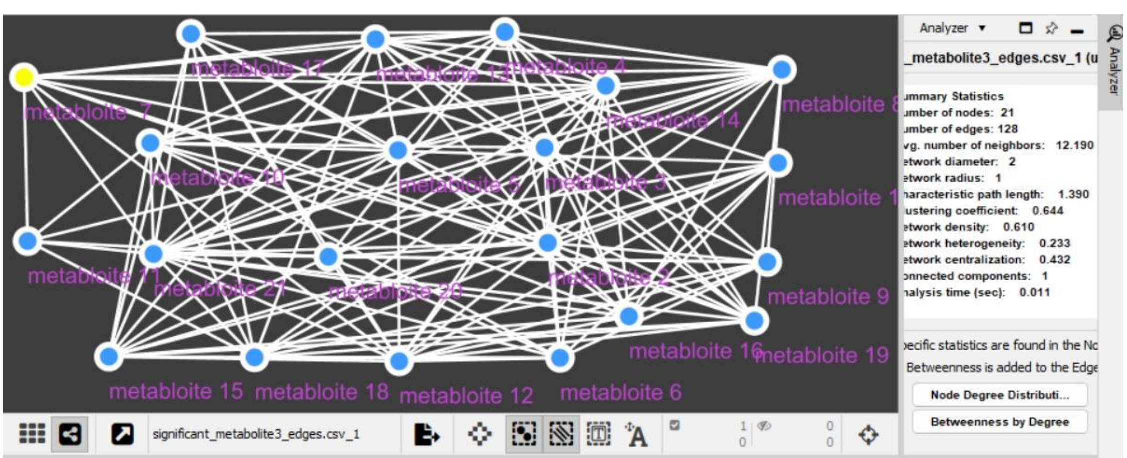
Task 5: Metabolic Networks

```
library(GeneNet)

# Assume corrected_matrix includes scanID in first column; remove it for
GeneNet input
corrected_data <- corrected_matrix[, -1]  # Remove scanID column
corrected_data <- as.matrix(corrected_data)  # Ensure matrix format

# Step 1: Estimate the shrinkage partial correlation matrix
pcor_matrix <- ggm.estimate.pcor(corrected_data)

## Warning: 1 instances of variables with zero scale detected!

## Estimating optimal shrinkage intensity lambda (correlation matrix): 0.9216

## Warning: 1 instances of variables with zero scale detected!

# Step 2: Test edges (returns a data frame of all possible edges)
network_edges <- network.test.edges(pcor_matrix)

## Estimate (local) false discovery rates (partial correlations):
## Step 1... determine cutoff point
## Step 2... estimate parameters of null distribution and eta0
## Step 3... compute p-values and estimate empirical PDF/CDF
```

```
## Step 4... compute q-values and local fdr
## Step 5... prepare for plotting
```




```r
network_edges$fdr <- p.adjust(network_edges$pval , method ="fdr")

significant_edges <- subset(network_edges,fdr>0.85)

# Step 4: Save the significant edges to a CSV file
write.csv(significant_edges, "significant_metabolite3_edges.csv", row.names =
FALSE)

knitr::include_graphics("task5.png")
```



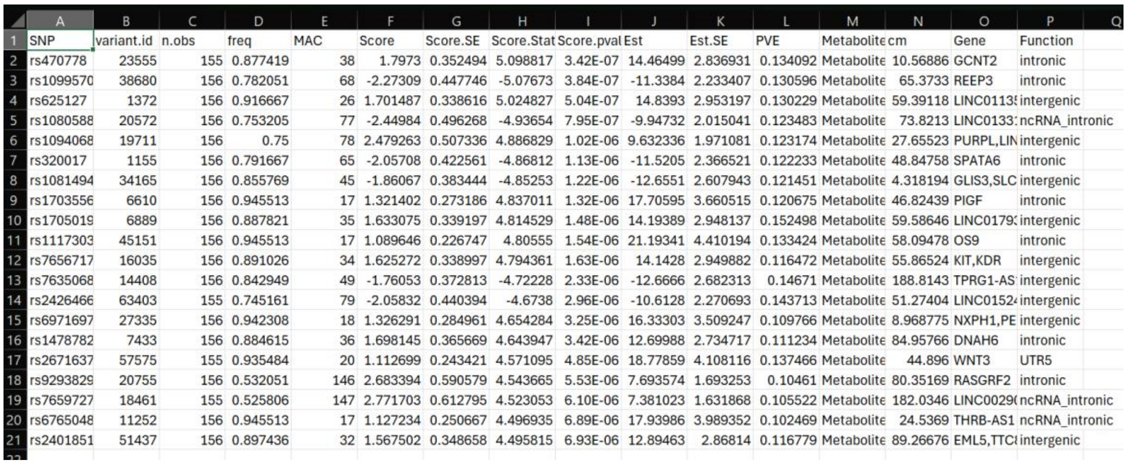
Task 6

```
# Read the map file
snp_map <- read.table("Qatari156_filtered_pruned.map", header = FALSE,
stringsAsFactors = FALSE)

# Assign proper column names
colnames(snp_map) <- c("chr", "SNP", "cm", "pos")

# Make sure chr and pos are the correct types
snp_map$chr <- as.character(snp_map$chr)
snp_map$pos <- as.numeric(snp_map$pos)

for (metab in names(significant_assoc_results)) {
  significant_assoc_results[[metab]] <-
merge(significant_assoc_results[[metab]], snp_map,
                                  by = c("chr", "pos"),
                                  all.x = TRUE)
}
final_sig_task6 <- do.call(rbind,significant_assoc_results)
write.xlsx(final_sig_task6, file = "Significant_mQTLs_snps.xlsx", rowNames =
FALSE)

# Read the Excel file
data <- read_xlsx("Significant_mQTLs_snps.xlsx")

# Process the data to get top 20 by p-value (keeping all columns)
top_20 <- data %>%
  filter(!is.na(Score.pval), !is.na(SNP), !is.na(pos)) %>%
  arrange(Score.pval) %>%  # Sort by p-value (ascending)
  slice_head(n = 20) %>%
  mutate(cm = round(cm, 6))  # Optionally round genetic distance

# View the result
print(top_20)

## # A tibble: 20 × 16
##    chr          pos variant.id n.obs  freq   MAC Score Score.SE Score.Stat
##    <chr>      <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>    <dbl>      <dbl>
##  1 6       10568858      23555   155 0.877    38  1.80    0.352       5.10
##  2 10      65373297      38680   156 0.782    68 -2.27    0.448      -5.08
##  3 1       59391181       1372   156 0.917    26  1.70    0.339       5.02
##  4 5       73821303      20572   156 0.753    77 -2.45    0.496      -4.94
##  5 5       27655227      19711   156 0.75     78  2.48    0.507       4.89
##  6 1       48847582       1155   156 0.792    65 -2.06    0.423      -4.87
##  7 9        4318194      34165   156 0.856    45 -1.86    0.383      -4.85
##  8 2       46824391       6610   156 0.946    17  1.32    0.273       4.84
##  9 2       59586461       6889   156 0.888    35  1.63    0.339       4.81
## 10 12      58094775      45151   156 0.946    17  1.09    0.227       4.81
## 11 4       55865237      16035   156 0.891    34  1.63    0.339       4.79
## 12 3      188814284      14408   156 0.843    49 -1.76    0.373      -4.72
## 13 20      51274041      63403   155 0.745    79 -2.06    0.440      -4.67
```

```
## 14 7       8968775      27335   156 0.942    18  1.33    0.285        4.65
## 15 2      84957658       7433   156 0.885    36  1.70    0.366        4.64
## 16 17     44896002      57575   155 0.935    20  1.11    0.243        4.57
## 17 5      80351685      20755   156 0.532   146  2.68    0.591        4.54
## 18 4     182034590      18461   155 0.526   147  2.77    0.613        4.52
## 19 3      24536898      11252   156 0.946    17  1.13    0.251        4.50
## 20 14     89266760      51437   156 0.897    32  1.57    0.349        4.50
## # ℹ 7 more variables: Score.pval <dbl>, Est <dbl>, Est.SE <dbl>, PVE
<dbl>,
## #   Metabolite <chr>, SNP <chr>, cm <dbl>

# Save to CSV
write.csv(top_20, "top_20_significant_snps.csv", row.names = FALSE)

library(dplyr)
library(readr)

# Load and prepare top 20 SNPs file
top20_df <- read_csv("top_20_significant_snps.csv", show_col_types = FALSE)
%>%
  mutate(
    chr = gsub("^chr", "", as.character(chr)),  # Clean 'chr' prefix if
present
    pos = as.integer(pos)
  )

# Load and prepare ANNOVAR annotation file
annovar_df <- read_tsv("annotated_output_ali.hg19_multianno.txt",
show_col_types = FALSE) %>%
  mutate(
    Chr = gsub("^chr", "", as.character(Chr)),  # Ensure consistency with
top20_df
    Start = as.integer(Start)
  )

# Select relevant annotation columns
annovar_subset <- annovar_df %>%
  select(Chr, Start, Gene = Gene.refGene, Function = Func.refGene)

# Annotate top SNPs using a left join
top20_annotated <- top20_df %>%
  left_join(annovar_subset, by = c("chr" = "Chr", "pos" = "Start"))

# Rearrange columns and remove temporary chr/pos
top20_annotated <- top20_annotated %>%
  select(SNP, everything(), -chr, -pos)

# Save to CSV
write_csv(top20_annotated, "top20_snps_with_gene_annotation_annovar.csv")
```

```r
# View in RStudio
View(top20_annotated)
```

```r
knitr::include_graphics("Screenshot 2025-05-20 022816.png")
```

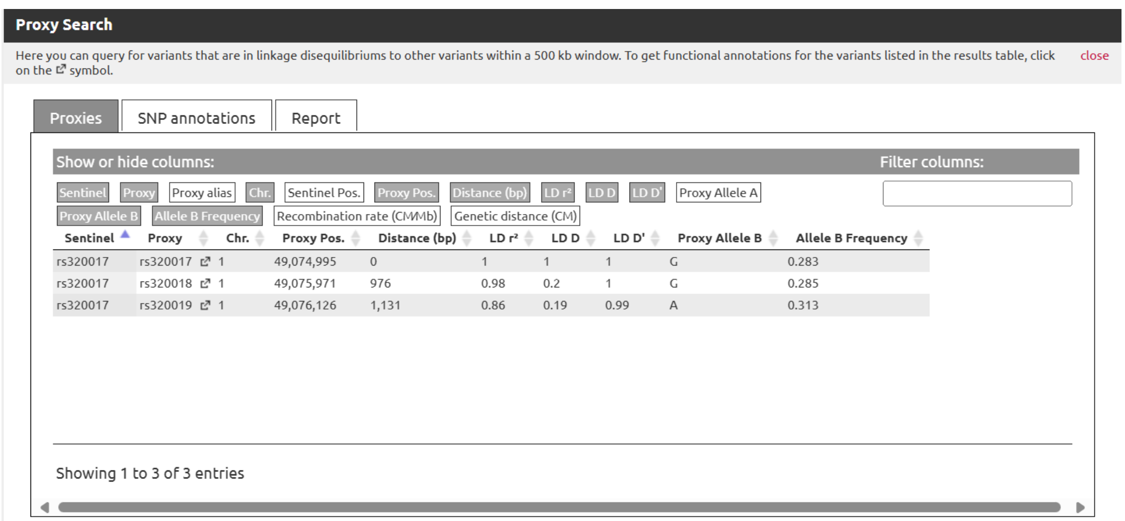


	SNP	variant.id	n.obs	freq	MAC	Score	Score.SE	Score.Stat	Score.pval	Est	Est.SE	PVE	Metabolite	cm	Gene	Function
2	rs470778	23555	155	0.877419	38	1.7973	0.352494	5.098817	3.42E-07	14.46499	2.836931	0.134092	Metabolite	10.56886	GCNT2	intronic
3	rs1099570	38680	156	0.782051	68	-2.27309	0.447746	-5.07673	3.84E-07	-11.3384	2.233407	0.130596	Metabolite	65.3733	REEP3	intronic
4	rs625127	1372	156	0.916667	26	1.701487	0.338616	5.024827	5.04E-07	14.8393	2.953197	0.130229	Metabolite	59.39118	LINC0113	intergenic
5	rs1080588	20572	156	0.753205	77	-2.44984	0.496268	-4.93654	7.95E-07	-9.94732	2.015041	0.123483	Metabolite	73.8213	LINC0133	ncRNA_intronic
6	rs1094068	19711	156	0.75	78	2.479263	0.507336	4.886829	1.02E-06	9.632336	1.971081	0.123174	Metabolite	27.65523	PURPL,LIN	intergenic
7	rs320017	1155	156	0.791667	65	-2.05708	0.422561	-4.86812	1.13E-06	-11.5205	2.366521	0.122233	Metabolite	48.84758	SPATA6	intronic
8	rs1081494	34165	156	0.855769	45	-1.86067	0.383444	-4.85253	1.22E-06	-12.6551	2.607943	0.121451	Metabolite	4.318194	GLIS3,SLC	intergenic
9	rs1703556	6610	156	0.945513	17	1.321402	0.273186	4.837011	1.32E-06	17.70595	3.660515	0.120675	Metabolite	46.82439	PIGF	intronic
10	rs1705019	6889	156	0.887821	35	1.633075	0.339197	4.814529	1.48E-06	14.19389	2.948137	0.152498	Metabolite	59.58646	LINC0179	intergenic
11	rs1117303	45151	156	0.945513	17	1.089646	0.226747	4.80555	1.54E-06	21.19341	4.410194	0.133424	Metabolite	58.09478	OS9	intronic
12	rs7656717	16035	156	0.891026	34	1.625272	0.338997	4.794361	1.63E-06	14.1428	2.949882	0.116472	Metabolite	55.86524	KIT,KDR	intergenic
13	rs7635068	14408	156	0.842949	49	-1.76053	0.372813	-4.72228	2.33E-06	-12.6666	2.682313	0.14671	Metabolite	188.8143	TPRG1-AS	intergenic
14	rs2426466	63403	155	0.745161	79	-2.05832	0.440394	-4.6738	2.96E-06	-10.6128	2.270693	0.143713	Metabolite	51.27404	LINC0152	intergenic
15	rs6971697	27335	156	0.942308	18	1.326291	0.284961	4.654284	3.25E-06	16.33303	3.509247	0.109766	Metabolite	8.968775	NXPH1,PE	intergenic
16	rs1478782	7433	156	0.884615	36	1.698145	0.365669	4.643947	3.42E-06	12.69988	2.734717	0.111234	Metabolite	84.95766	DNAH6	intronic
17	rs2671637	57575	155	0.935484	20	1.112699	0.243421	4.571095	4.85E-06	18.77859	4.108116	0.137466	Metabolite	44.896	WNT3	UTR5
18	rs9293829	20755	156	0.532051	146	2.683394	0.590579	4.543665	5.53E-06	7.693574	1.693253	0.10461	Metabolite	80.35169	RASGRF2	intronic
19	rs7659727	18461	155	0.525806	147	2.771703	0.612795	4.523053	6.10E-06	7.381023	1.631868	0.105522	Metabolite	182.0346	LINC0029	ncRNA_intronic
20	rs6765048	11252	156	0.945513	17	1.127234	0.250667	4.496935	6.89E-06	17.93986	3.989352	0.102469	Metabolite	24.5369	THRB-AS1	ncRNA_intronic
21	rs2401851	51437	156	0.897436	32	1.567502	0.348658	4.495815	6.93E-06	12.89463	2.86814	0.116779	Metabolite	89.26676	EML5,TTC	intergenic

Task 7

for rs320017

```r
knitr::include_graphics("Screenshot 2025-05-19 164007.png")
```



Proxy Search

Here you can query for variants that are in linkage disequilibriums to other variants within a 500 kb window. To get functional annotations for the variants listed in the results table, click on the symbol. close

Proxies | SNP annotations | Report

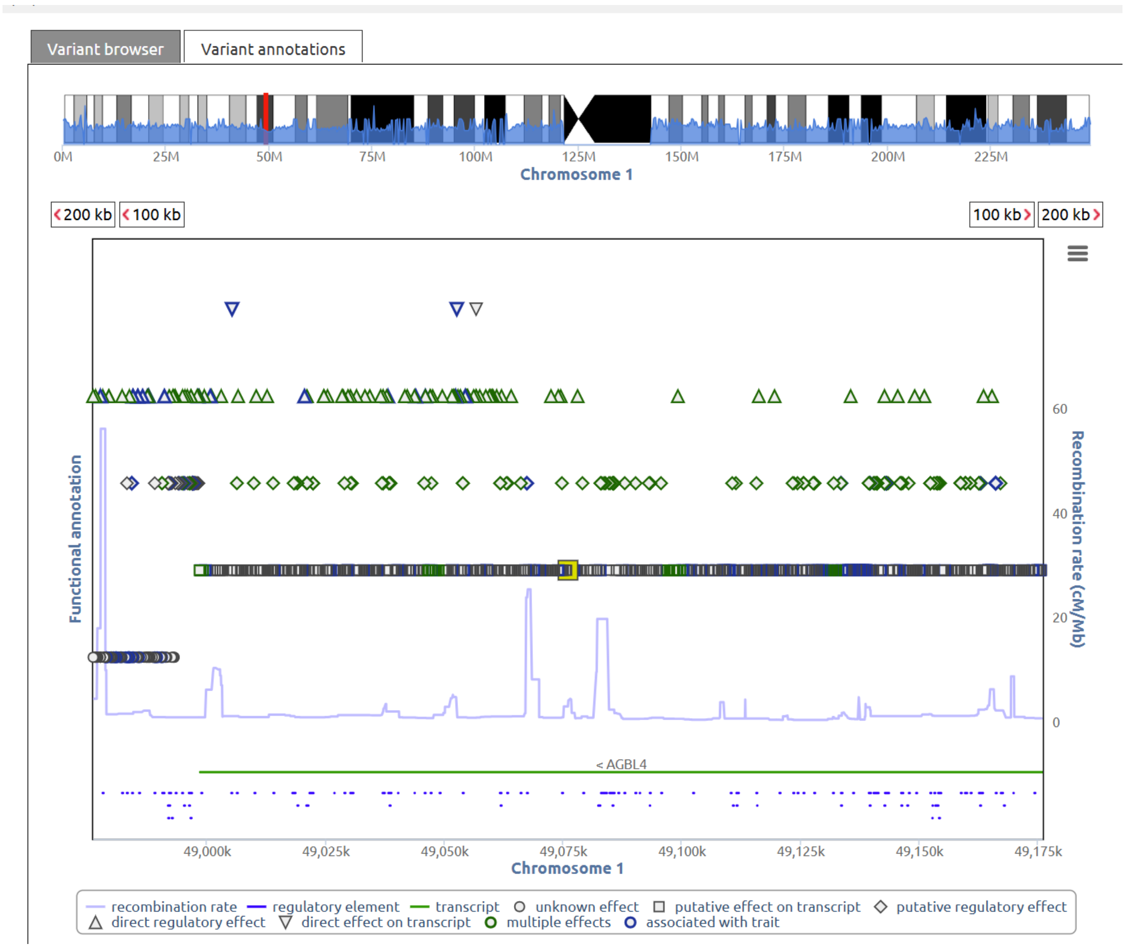
Show or hide columns: Sentinel, Proxy, Proxy alias, Chr., Sentinel Pos., Proxy Pos., Distance (bp), LD r², LD D, LD D', Proxy Allele A, Proxy Allele B, Allele B Frequency, Recombination rate (CM/Mb), Genetic distance (CM)

Filter columns:

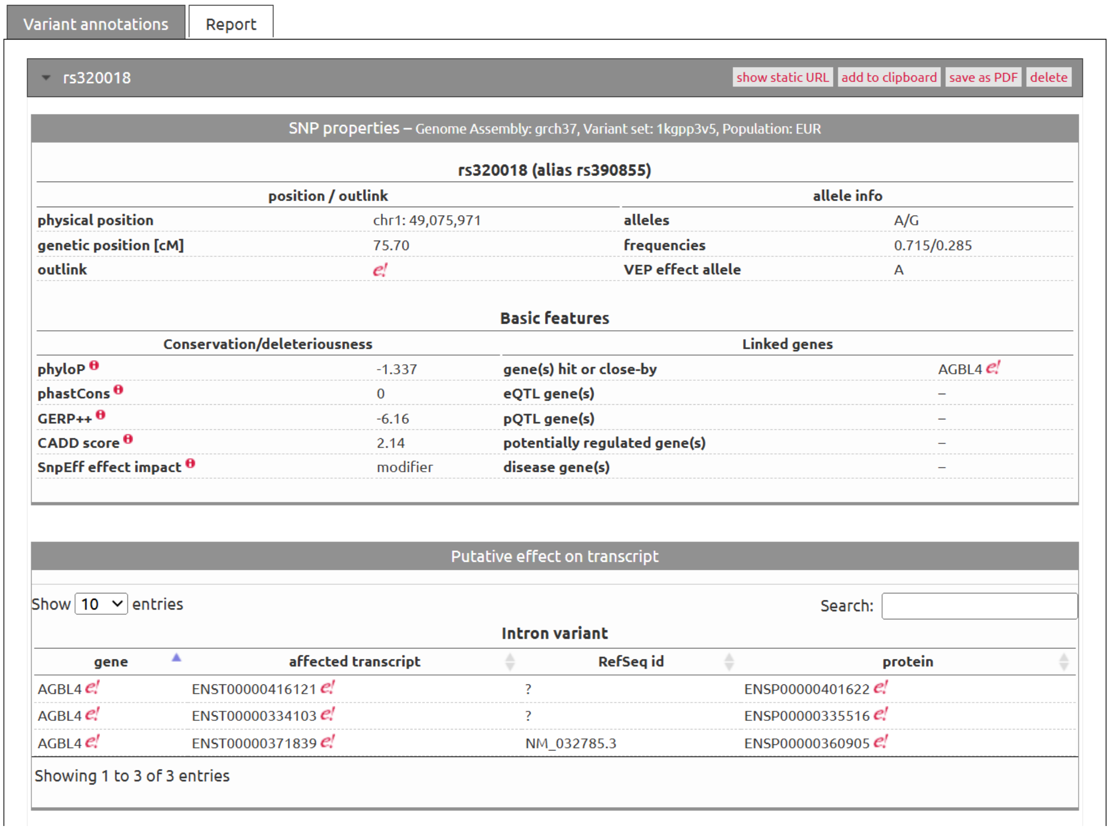
Sentinel	Proxy	Chr.	Proxy Pos.	Distance (bp)	LD r²	LD D	LD D'	Proxy Allele B	Allele B Frequency
rs320017	rs320017	1	49,074,995	0	1	1	1	G	0.283
rs320017	rs320018	1	49,075,971	976	0.98	0.2	1	G	0.285
rs320017	rs320019	1	49,076,126	1,131	0.86	0.19	0.99	A	0.313

Showing 1 to 3 of 3 entries

```r
knitr::include_graphics("Screenshot 2025-05-20 000704.png")
```




```
knitr::include_graphics("Screenshot 2025-05-20 020554.png")
```



Variant annotations | Report

rs320018 — show static URL | add to clipboard | save as PDF | delete

SNP properties – Genome Assembly: grch37, Variant set: 1kgpp3v5, Population: EUR

rs320018 (alias rs390855)

position / outlink		allele info	
physical position	chr1: 49,075,971	alleles	A/G
genetic position [cM]	75.70	frequencies	0.715/0.285
outlink	e!	VEP effect allele	A

Basic features

Conservation/deleteriousness		Linked genes	
phyloP	-1.337	gene(s) hit or close-by	AGBL4 e!
phastCons	0	eQTL gene(s)	–
GERP++	-6.16	pQTL gene(s)	–
CADD score	2.14	potentially regulated gene(s)	–
SnpEff effect impact	modifier	disease gene(s)	–

Putative effect on transcript

Show 10 entries — Search:

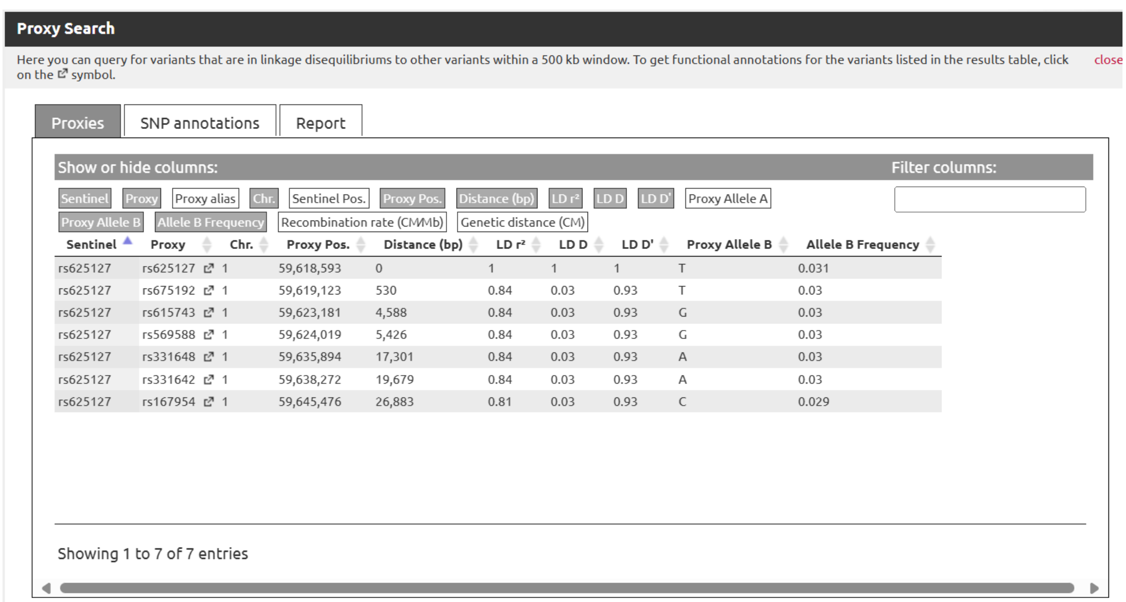
Intron variant

gene	affected transcript	RefSeq id	protein
AGBL4 e!	ENST00000416121 e!	?	ENSP00000401622 e!
AGBL4 e!	ENST00000334103 e!	?	ENSP00000335516 e!
AGBL4 e!	ENST00000371839 e!	NM_032785.3	ENSP00000360905 e!

Showing 1 to 3 of 3 entries

for rs625127

```r
knitr::include_graphics("Screenshot 2025-05-19 164135.png")
```



Proxy Search

Here you can query for variants that are in linkage disequilibriums to other variants within a 500 kb window. To get functional annotations for the variants listed in the results table, click on the symbol. — close

Proxies | SNP annotations | Report

Show or hide columns: Sentinel | Proxy | Proxy alias | Chr. | Sentinel Pos. | Proxy Pos. | Distance (bp) | LD r² | LD D | LD D' | Proxy Allele A | Proxy Allele B | Allele B Frequency | Recombination rate (CMMb) | Genetic distance (CM)

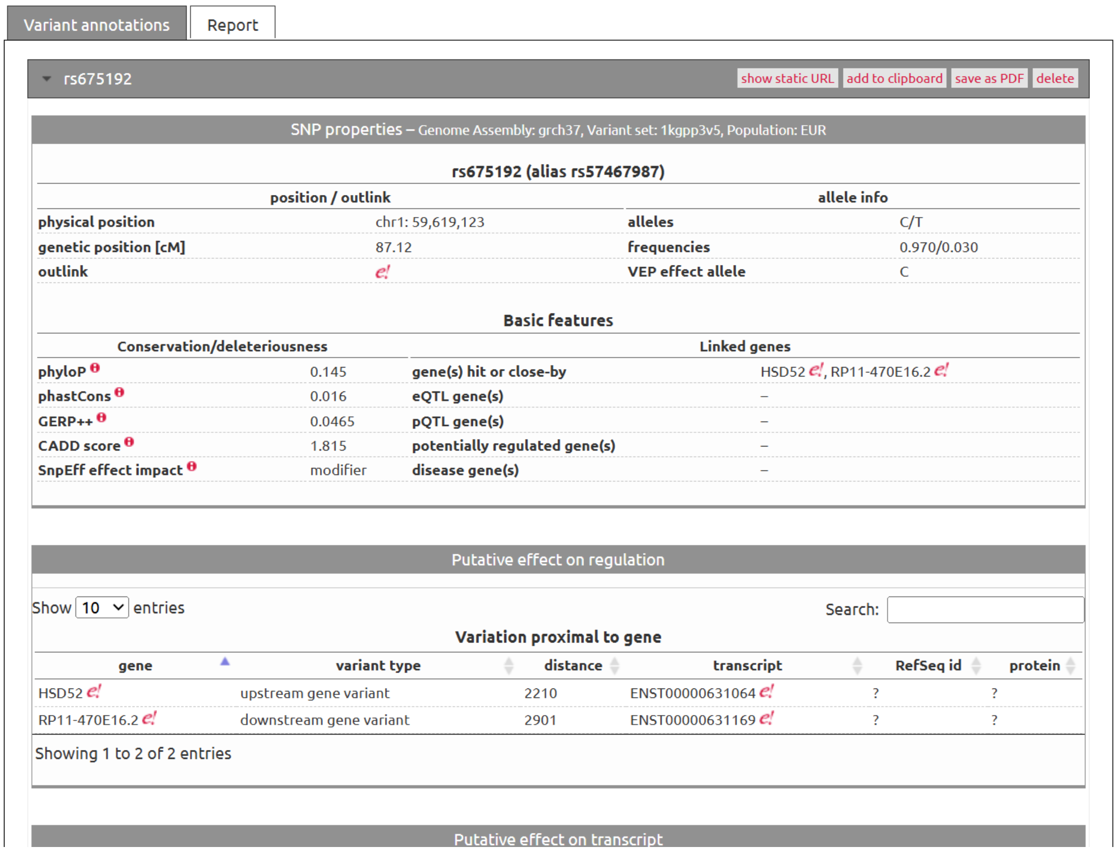
Filter columns:

Sentinel	Proxy	Chr.	Proxy Pos.	Distance (bp)	LD r²	LD D	LD D'	Proxy Allele B	Allele B Frequency
rs625127	rs625127	1	59,618,593	0	1	1	1	T	0.031
rs625127	rs675192	1	59,619,123	530	0.84	0.03	0.93	T	0.03
rs625127	rs615743	1	59,623,181	4,588	0.84	0.03	0.93	G	0.03
rs625127	rs569588	1	59,624,019	5,426	0.84	0.03	0.93	G	0.03
rs625127	rs331648	1	59,635,894	17,301	0.84	0.03	0.93	A	0.03
rs625127	rs331642	1	59,638,272	19,679	0.84	0.03	0.93	A	0.03
rs625127	rs167954	1	59,645,476	26,883	0.81	0.03	0.93	C	0.029

Showing 1 to 7 of 7 entries

```r
knitr::include_graphics("Screenshot 2025-05-20 001019.png")
```


```r
knitr::include_graphics("Screenshot 2025-05-20 020755.png")
```



Variant annotations | Report

rs675192 — show static URL | add to clipboard | save as PDF | delete

SNP properties – Genome Assembly: grch37, Variant set: 1kgpp3v5, Population: EUR

rs675192 (alias rs57467987)

position / outlink		allele info	
physical position	chr1: 59,619,123	alleles	C/T
genetic position [cM]	87.12	frequencies	0.970/0.030
outlink		VEP effect allele	C

Basic features

Conservation/deleteriousness		Linked genes	
phyloP	0.145	gene(s) hit or close-by	HSD52, RP11-470E16.2
phastCons	0.016	eQTL gene(s)	–
GERP++	0.0465	pQTL gene(s)	–
CADD score	1.815	potentially regulated gene(s)	–
SnpEff effect impact	modifier	disease gene(s)	–

Putative effect on regulation

Show 10 entries — Search:

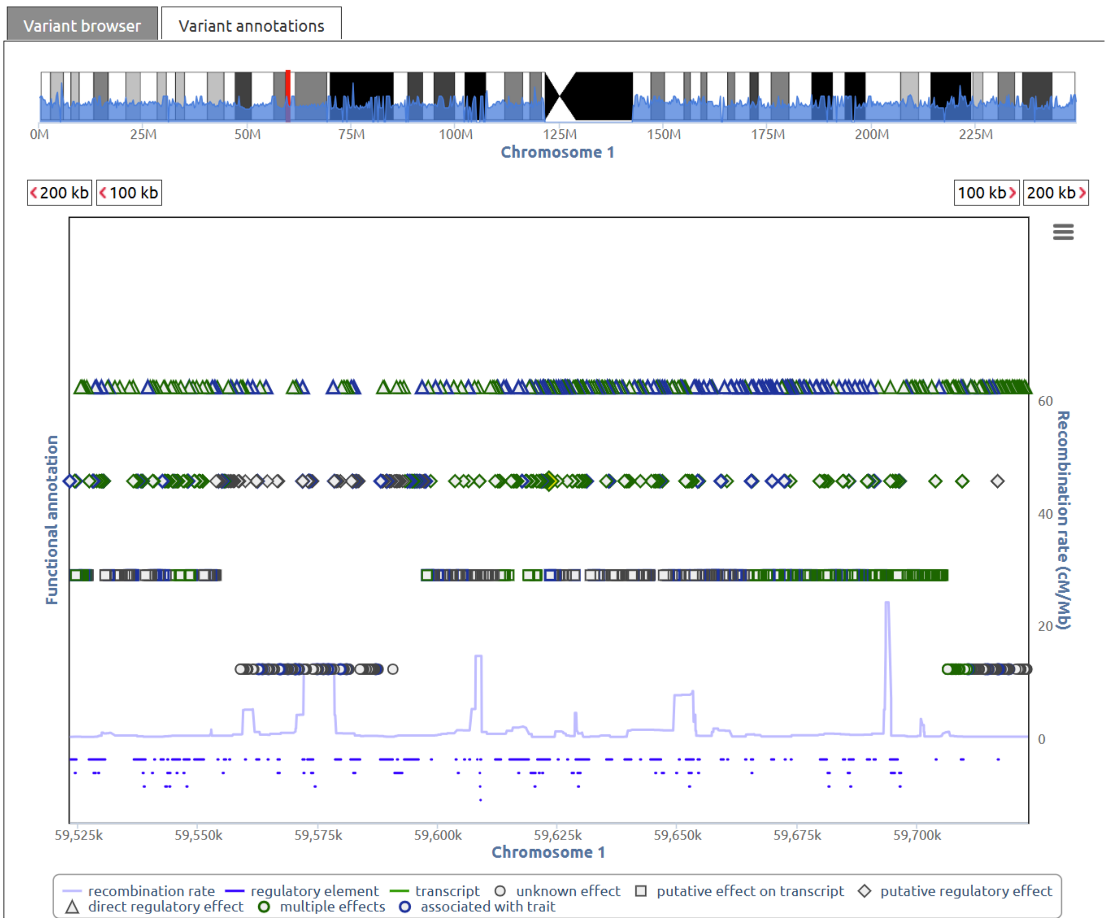
Variation proximal to gene

gene	variant type	distance	transcript	RefSeq id	protein
HSD52	upstream gene variant	2210	ENST00000631064	?	?
RP11-470E16.2	downstream gene variant	2901	ENST00000631169	?	?

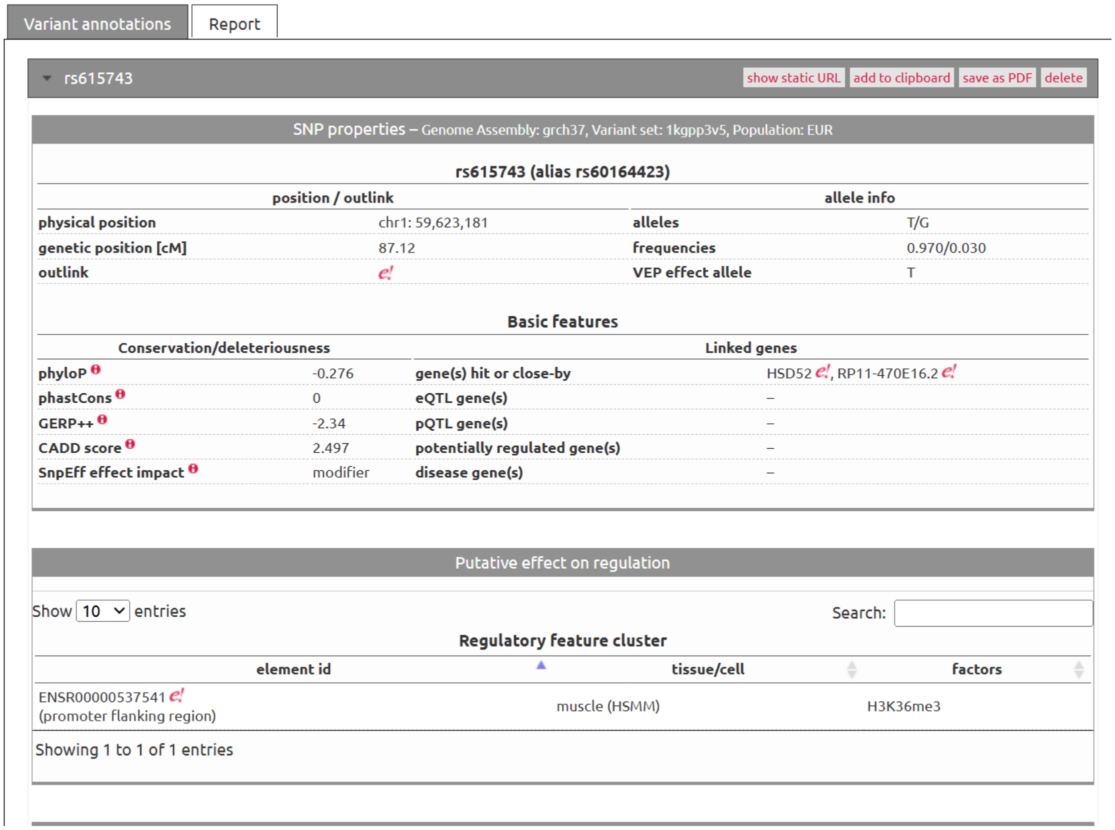
Showing 1 to 2 of 2 entries

Putative effect on transcript

```r
knitr::include_graphics("Screenshot 2025-05-20 001210.png")
```




```r
knitr::include_graphics("Screenshot 2025-05-20 020901.png")
```



Variant annotations | Report

rs615743 — show static URL | add to clipboard | save as PDF | delete

SNP properties – Genome Assembly: grch37, Variant set: 1kgpp3v5, Population: EUR

rs615743 (alias rs60164423)

position / outlink		allele info	
physical position	chr1: 59,623,181	alleles	T/G
genetic position [cM]	87.12	frequencies	0.970/0.030
outlink		VEP effect allele	T

Basic features

Conservation/deleteriousness		Linked genes	
phyloP	-0.276	gene(s) hit or close-by	HSD52, RP11-470E16.2
phastCons	0	eQTL gene(s)	–
GERP++	-2.34	pQTL gene(s)	–
CADD score	2.497	potentially regulated gene(s)	–
SnpEff effect impact	modifier	disease gene(s)	–

Putative effect on regulation

Show 10 entries — Search:

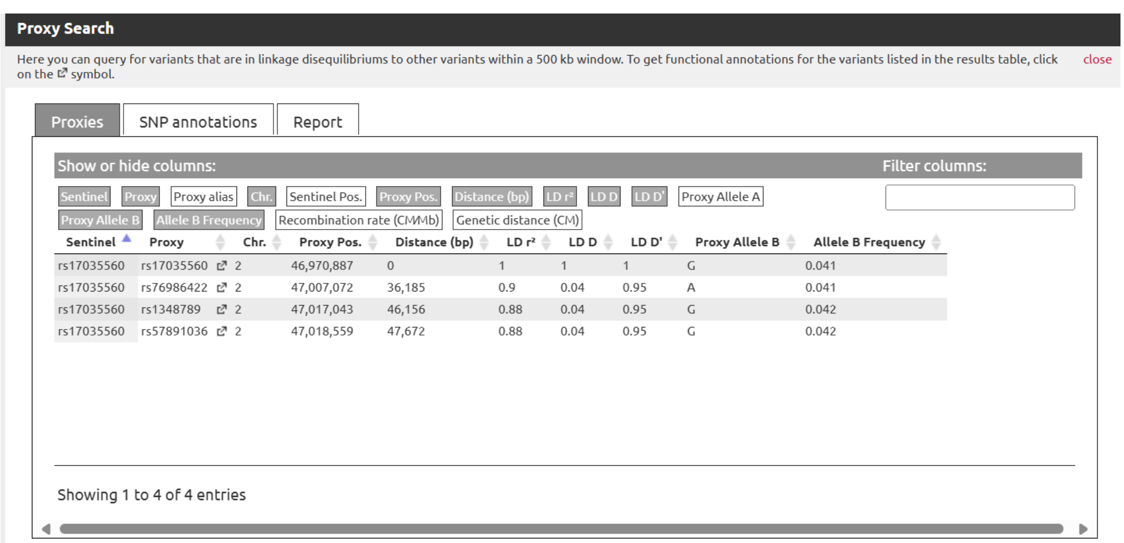
Regulatory feature cluster

element id	tissue/cell	factors
ENSR00000537541 (promoter flanking region)	muscle (HSMM)	H3K36me3

Showing 1 to 1 of 1 entries

for rs17035560

```r
knitr::include_graphics("Screenshot 2025-05-19 164236.png")
```



Proxy Search

Here you can query for variants that are in linkage disequilibriums to other variants within a 500 kb window. To get functional annotations for the variants listed in the results table, click on the symbol. close

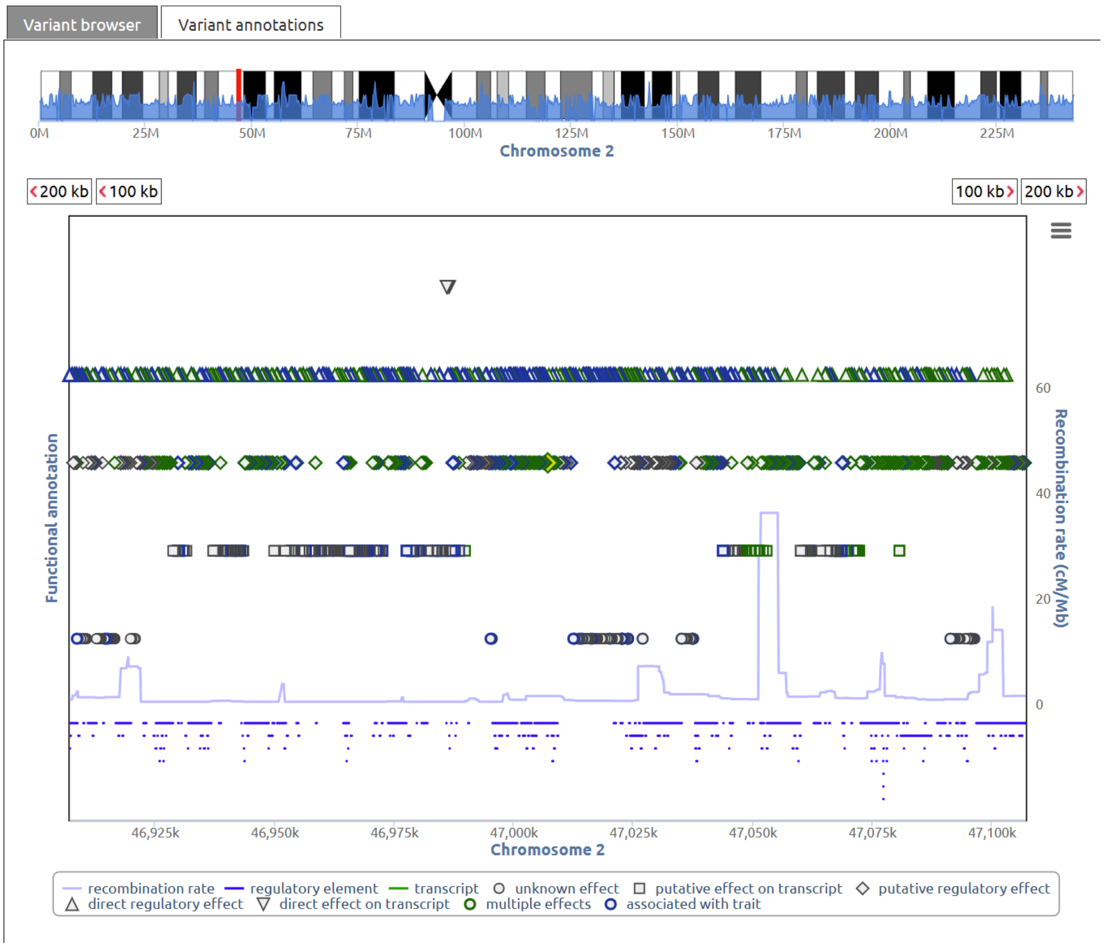
Proxies | SNP annotations | Report

Show or hide columns: Sentinel, Proxy, Proxy alias, Chr., Sentinel Pos., Proxy Pos., Distance (bp), LD r², LD D, LD D', Proxy Allele A, Proxy Allele B, Allele B Frequency, Recombination rate (CM/Mb), Genetic distance (CM) — Filter columns:

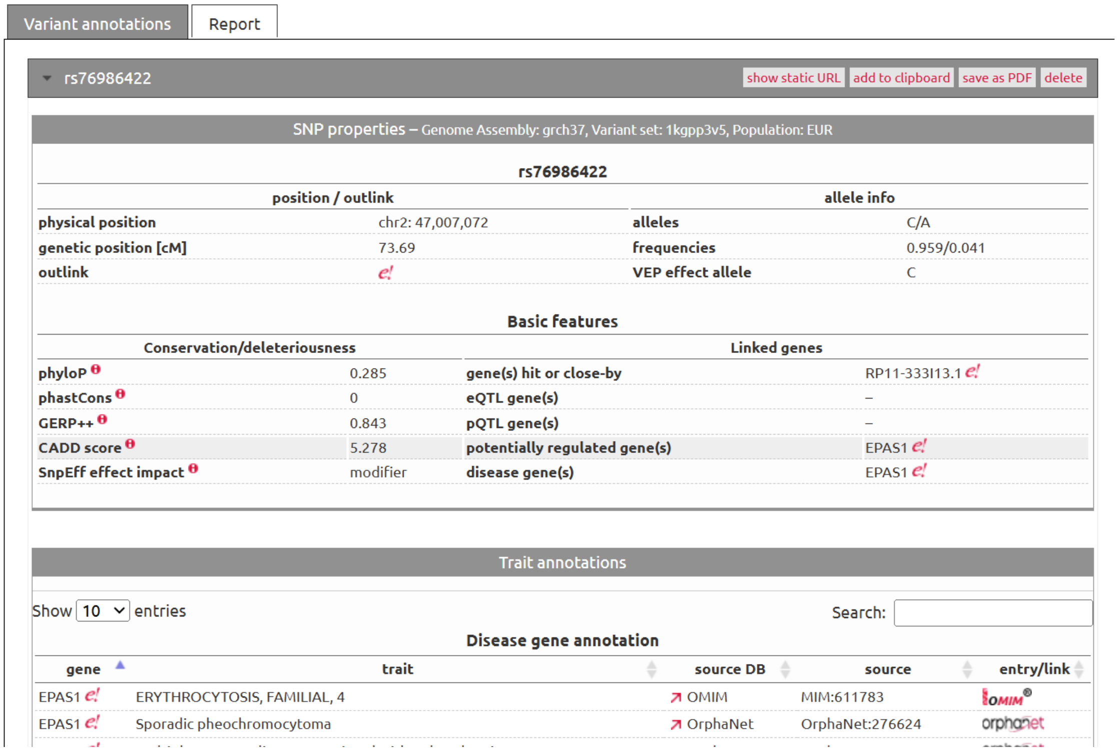
Sentinel	Proxy	Chr.	Proxy Pos.	Distance (bp)	LD r²	LD D	LD D'	Proxy Allele B	Allele B Frequency
rs17035560	rs17035560	2	46,970,887	0	1	1	1	G	0.041
rs17035560	rs76986422	2	47,007,072	36,185	0.9	0.04	0.95	A	0.041
rs17035560	rs1348789	2	47,017,043	46,156	0.88	0.04	0.95	G	0.042
rs17035560	rs57891036	2	47,018,559	47,672	0.88	0.04	0.95	G	0.042

Showing 1 to 4 of 4 entries

```r
knitr::include_graphics("Screenshot 2025-05-20 001408.png")
```




```r
knitr::include_graphics("Screenshot 2025-05-20 021010.png")
```



Variant annotations | Report

rs76986422 — show static URL | add to clipboard | save as PDF | delete

SNP properties – Genome Assembly: grch37, Variant set: 1kgpp3v5, Population: EUR

rs76986422

position / outlink		allele info	
physical position	chr2: 47,007,072	alleles	C/A
genetic position [cM]	73.69	frequencies	0.959/0.041
outlink	e!	VEP effect allele	C

Basic features

Conservation/deleteriousness		Linked genes	
phyloP	0.285	gene(s) hit or close-by	RP11-333I13.1 e!
phastCons	0	eQTL gene(s)	–
GERP++	0.843	pQTL gene(s)	–
CADD score	5.278	potentially regulated gene(s)	EPAS1 e!
SnpEff effect impact	modifier	disease gene(s)	EPAS1 e!

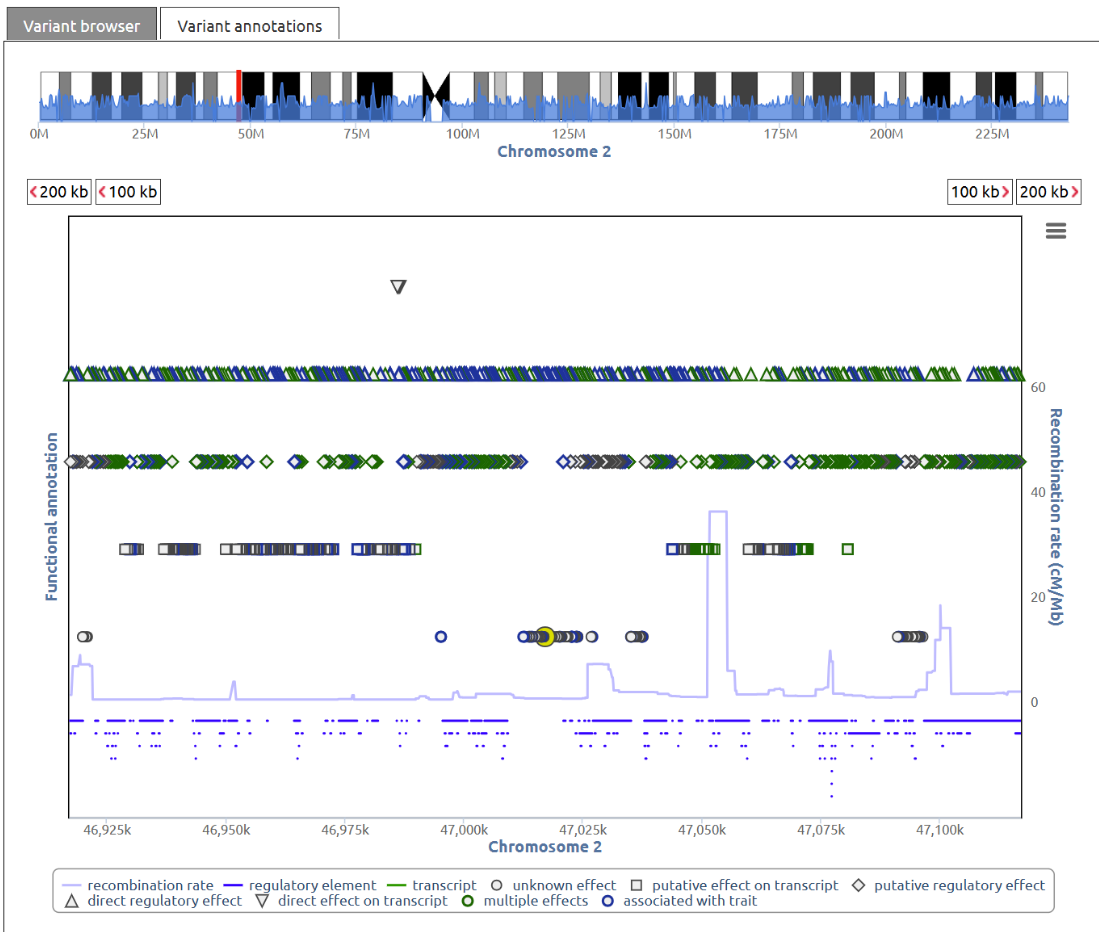
Trait annotations

Show 10 entries — Search:

Disease gene annotation

gene	trait	source DB	source	entry/link
EPAS1 e!	ERYTHROCYTOSIS, FAMILIAL, 4	OMIM	MIM:611783	OMIM
EPAS1 e!	Sporadic pheochromocytoma	OrphaNet	OrphaNet:276624	orphanet

```r
knitr::include_graphics("Screenshot 2025-05-20 001536.png")
```




```r
knitr::include_graphics("Screenshot 2025-05-20 021059.png")
```

Variant Annotation

This module allows you to get detailed annotations for one or more variants. If the results are not what you have expected, please check the "Report" tab for details.

Variant annotations | Report

rs1348789 — show static URL | add to clipboard | save as PDF | delete

SNP properties – Genome Assembly: grch37, Variant set: 1kgpp3v5, Population: EUR

rs1348789 (alias rs9284748, rs17496665)

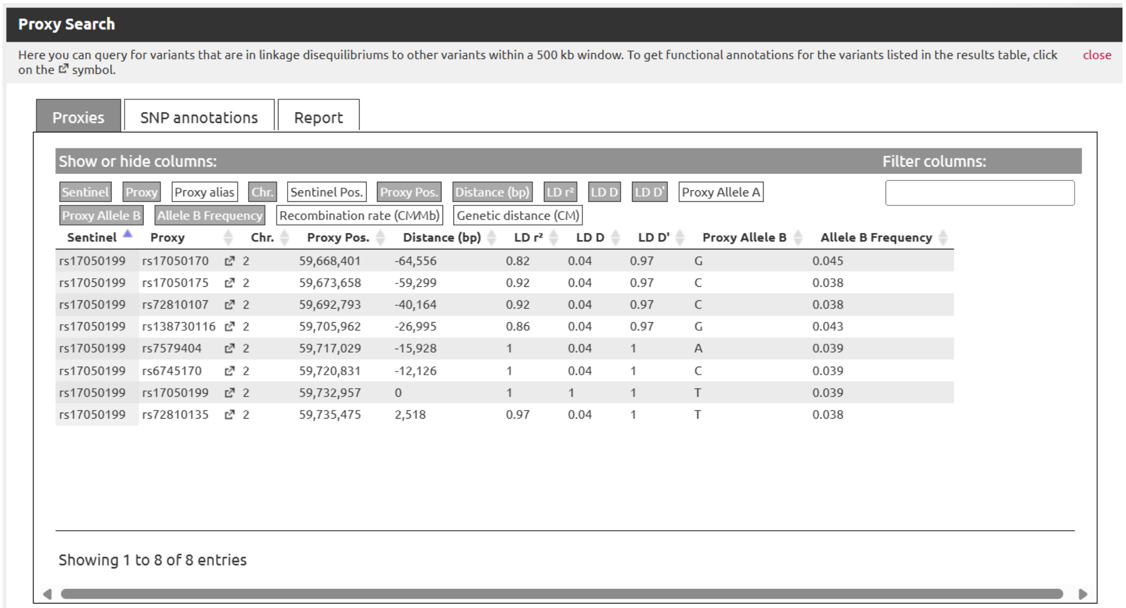
position / outlink		allele info	
physical position	chr2: 47,017,043	**alleles**	A/G
genetic position [cM]	73.70	**frequencies**	0.958/0.042
outlink	e!	**VEP effect allele**	A

Basic features

Conservation/deleteriousness		Linked genes	
phyloP	0.238	**gene(s) hit or close-by**	–
phastCons	0.102	**eQTL gene(s)**	–
GERP++	0.149	**pQTL gene(s)**	–
CADD score	2.98	**potentially regulated gene(s)**	–
SnpEff effect impact	modifier	**disease gene(s)**	–

for rs17050199

```r
knitr::include_graphics("Screenshot 2025-05-19 164505.png")
```



Proxy Search

Here you can query for variants that are in linkage disequilibriums to other variants within a 500 kb window. To get functional annotations for the variants listed in the results table, click on the symbol. close

Proxies | SNP annotations | Report

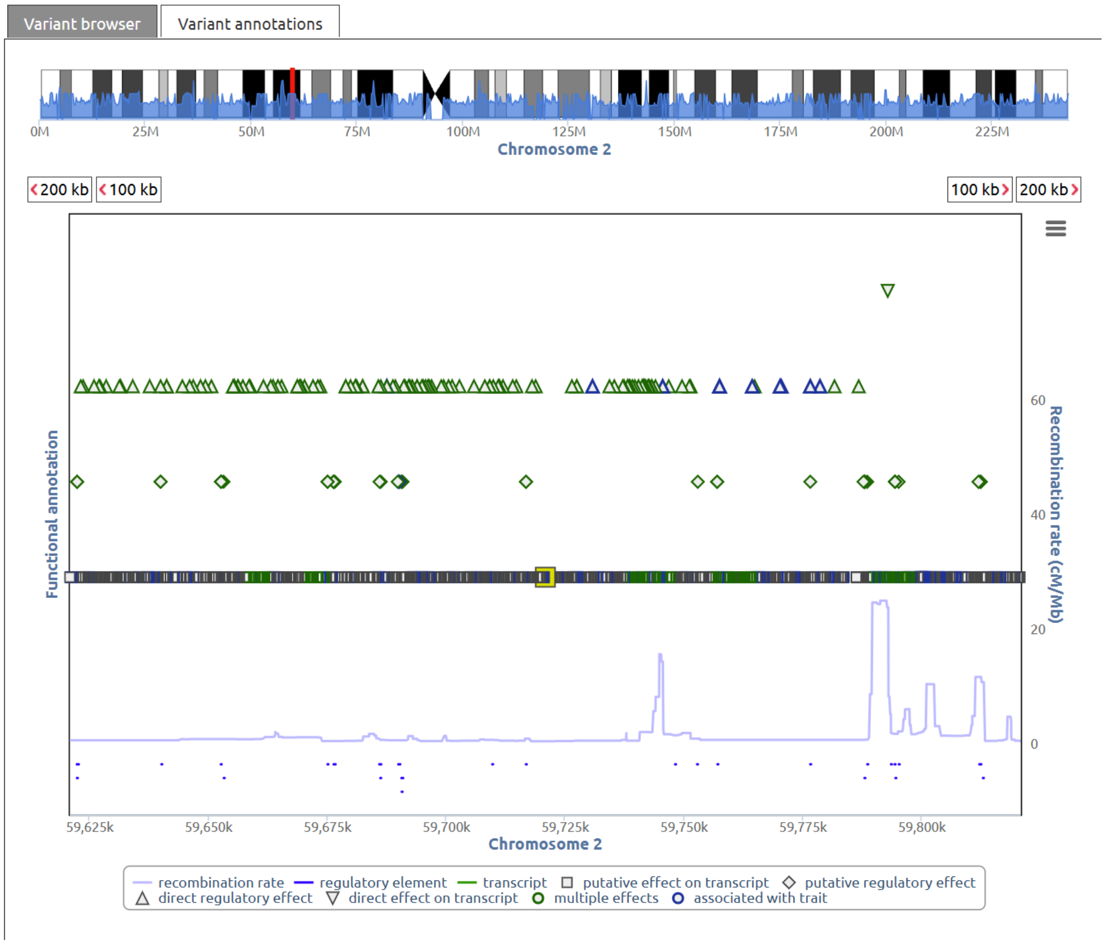
Show or hide columns: Sentinel | Proxy | Proxy alias | Chr. | Sentinel Pos. | Proxy Pos. | Distance (bp) | LD r² | LD D | LD D' | Proxy Allele A | Proxy Allele B | Allele B Frequency | Recombination rate (CM/Mb) | Genetic distance (CM)

Filter columns:

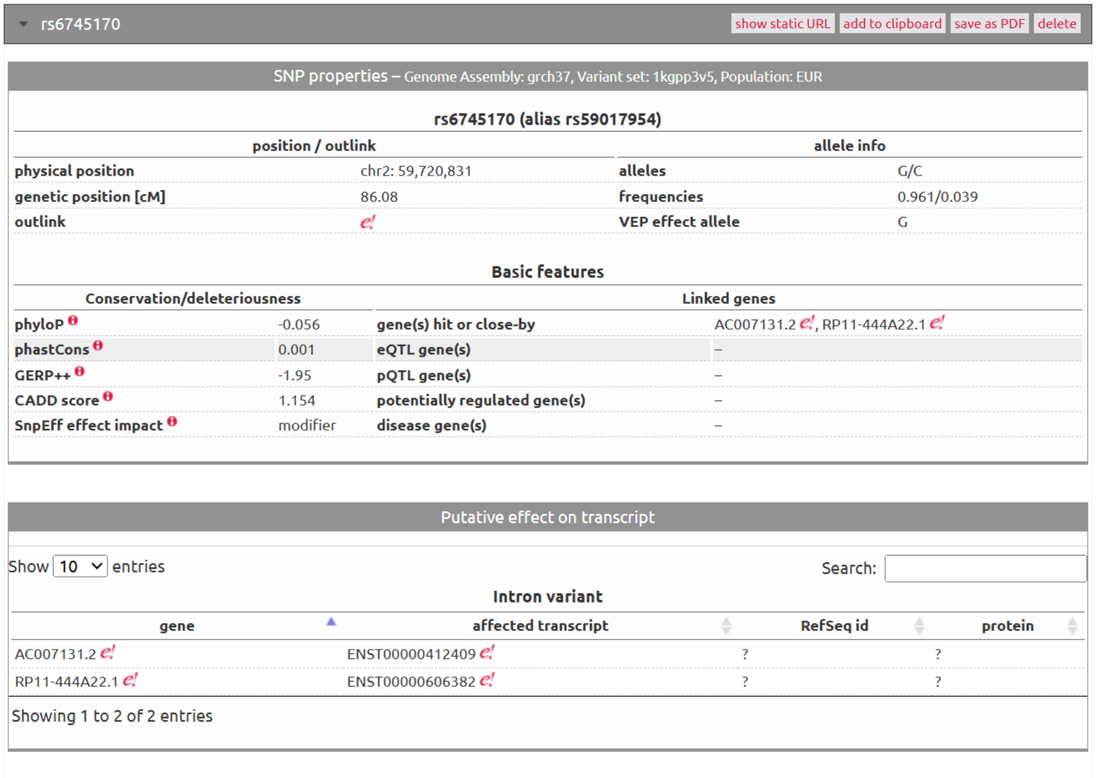
Sentinel	Proxy	Chr.	Proxy Pos.	Distance (bp)	LD r²	LD D	LD D'	Proxy Allele B	Allele B Frequency
rs17050199	rs17050170	2	59,668,401	-64,556	0.82	0.04	0.97	G	0.045
rs17050199	rs17050175	2	59,673,658	-59,299	0.92	0.04	0.97	C	0.038
rs17050199	rs72810107	2	59,692,793	-40,164	0.92	0.04	0.97	C	0.038
rs17050199	rs138730116	2	59,705,962	-26,995	0.86	0.04	0.97	G	0.043
rs17050199	rs7579404	2	59,717,029	-15,928	1	0.04	1	A	0.039
rs17050199	rs6745170	2	59,720,831	-12,126	1	0.04	1	C	0.039
rs17050199	rs17050199	2	59,732,957	0	1	1	1	T	0.039
rs17050199	rs72810135	2	59,735,475	2,518	0.97	0.04	1	T	0.038

Showing 1 to 8 of 8 entries

```r
knitr::include_graphics("Screenshot 2025-05-20 001825.png")
```




```r
knitr::include_graphics("Screenshot 2025-05-20 021323.png")
```



rs6745170 show static URL add to clipboard save as PDF delete

SNP properties – Genome Assembly: grch37, Variant set: 1kgpp3v5, Population: EUR

rs6745170 (alias rs59017954)

position / outlink		allele info	
physical position	chr2: 59,720,831	**alleles**	G/C
genetic position [cM]	86.08	**frequencies**	0.961/0.039
outlink	e!	**VEP effect allele**	G

Basic features

Conservation/deleteriousness		Linked genes	
phyloP	-0.056	**gene(s) hit or close-by**	AC007131.2 e!, RP11-444A22.1 e!
phastCons	0.001	**eQTL gene(s)**	–
GERP++	-1.95	**pQTL gene(s)**	–
CADD score	1.154	**potentially regulated gene(s)**	–
SnpEff effect impact	modifier	**disease gene(s)**	–

Putative effect on transcript

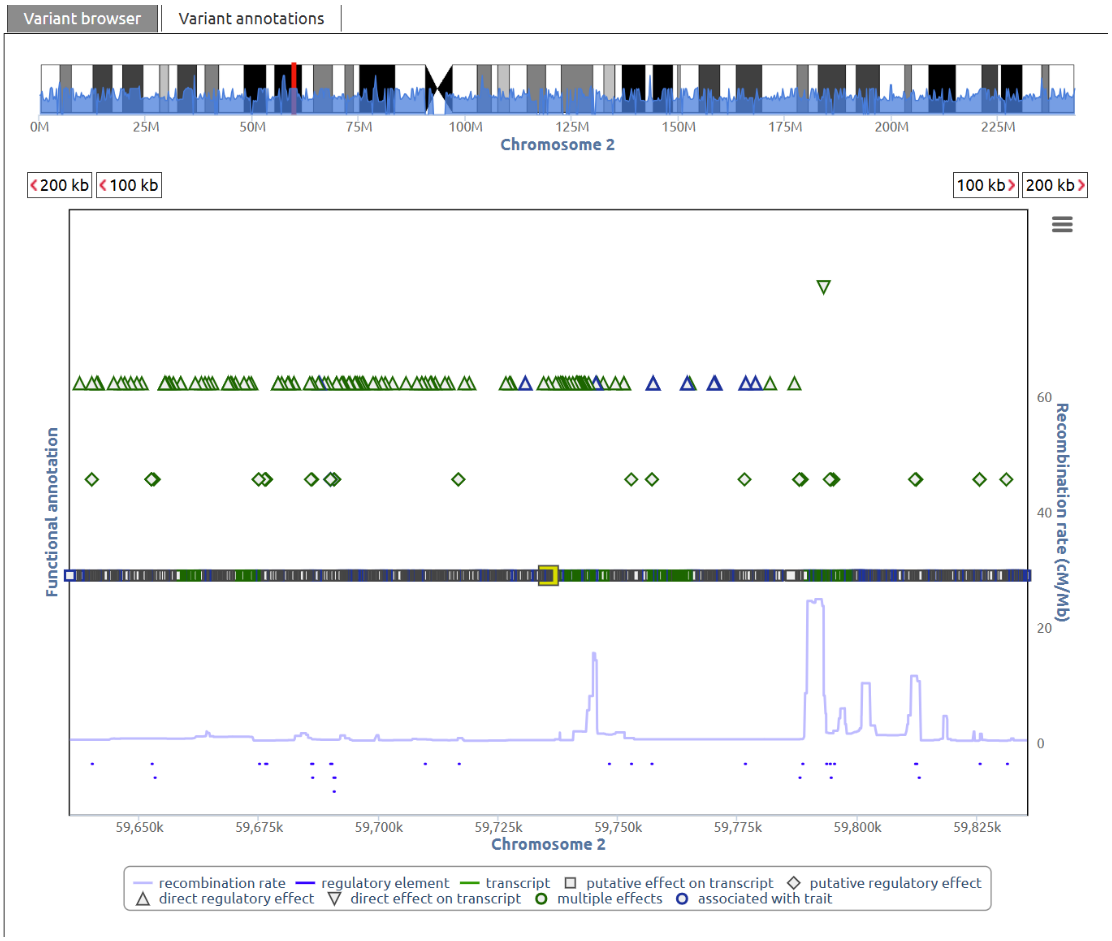
Show 10 entries Search:

Intron variant

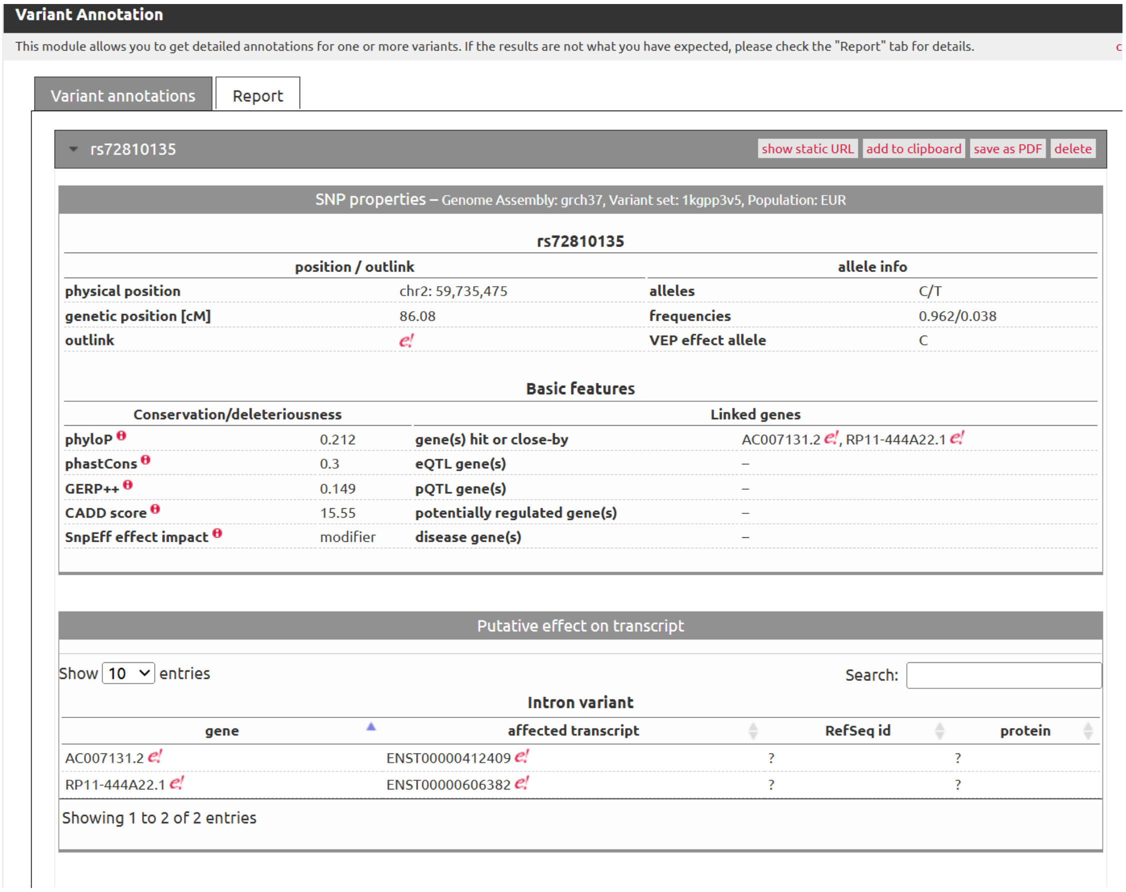
gene	affected transcript	RefSeq id	protein
AC007131.2 e!	ENST00000412409 e!	?	?
RP11-444A22.1 e!	ENST00000606382 e!	?	?

Showing 1 to 2 of 2 entries

```r
knitr::include_graphics("Screenshot 2025-05-20 001932.png")
```




```
knitr::include_graphics("Screenshot 2025-05-20 021407.png")
```



Variant Annotation

This module allows you to get detailed annotations for one or more variants. If the results are not what you have expected, please check the "Report" tab for details.

Variant annotations | Report

rs72810135 — show static URL | add to clipboard | save as PDF | delete

SNP properties – Genome Assembly: grch37, Variant set: 1kgpp3v5, Population: EUR

rs72810135

position / outlink		allele info	
physical position	chr2: 59,735,475	alleles	C/T
genetic position [cM]	86.08	frequencies	0.962/0.038
outlink	e!	VEP effect allele	C

Basic features

Conservation/deleteriousness		Linked genes	
phyloP	0.212	gene(s) hit or close-by	AC007131.2 e!, RP11-444A22.1 e!
phastCons	0.3	eQTL gene(s)	–
GERP++	0.149	pQTL gene(s)	–
CADD score	15.55	potentially regulated gene(s)	–
SnpEff effect impact	modifier	disease gene(s)	–

Putative effect on transcript

Show 10 entries Search:

Intron variant

gene	affected transcript	RefSeq id	protein
AC007131.2 e!	ENST00000412409 e!	?	?
RP11-444A22.1 e!	ENST00000606382 e!	?	?

Showing 1 to 2 of 2 entries

for rs1478782

```r
knitr::include_graphics("Screenshot 2025-05-19 164639.png")
```

Proxy Search

Here you can query for variants that are in linkage disequilibriums to other variants within a 500 kb window. To get functional annotations for the variants listed in the results table, click on the ↗ symbol. close

Proxies | SNP annotations | Report

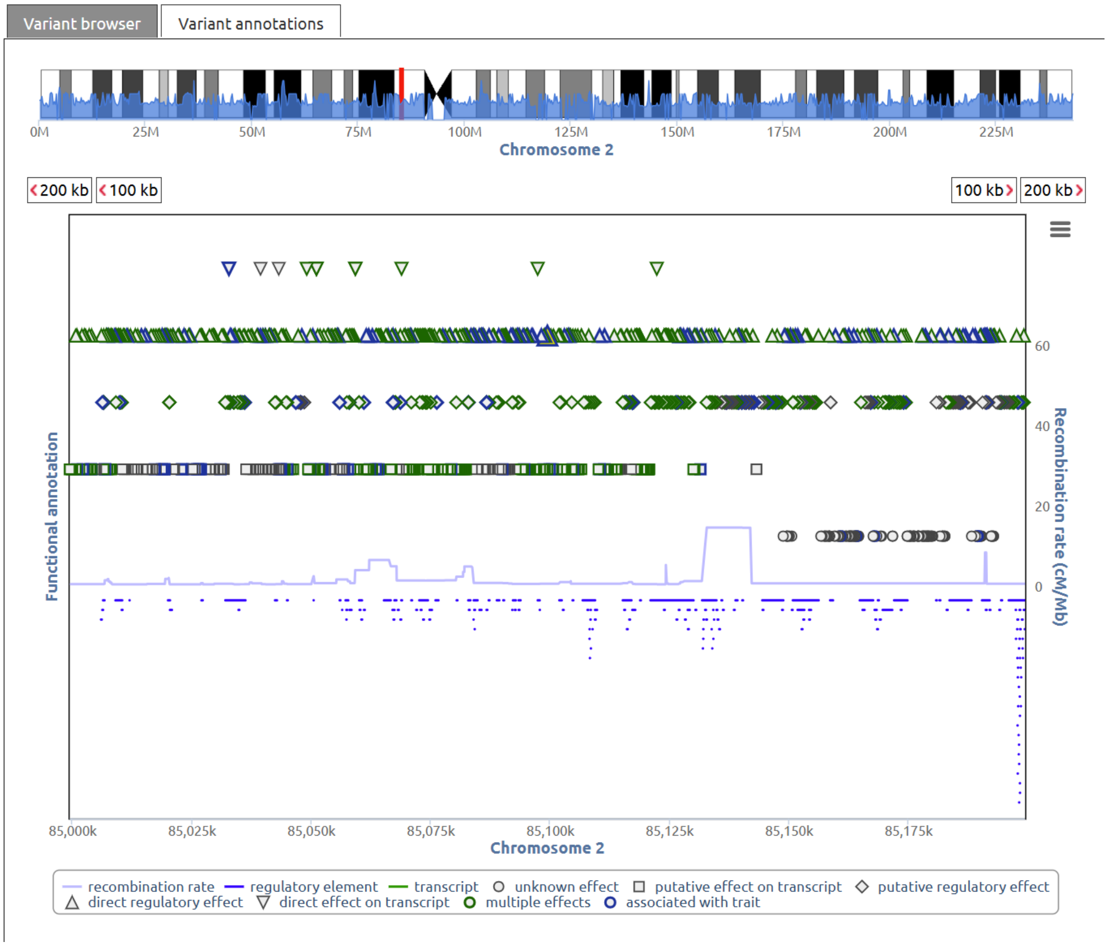
Show or hide columns: Filter columns:

Sentinel | Proxy | Proxy alias | Chr. | Sentinel Pos. | Proxy Pos. | Distance (bp) | LD r² | LD D | LD D' | Proxy Allele A | Proxy Allele B | Allele B Frequency | Recombination rate (CM/Mb) | Genetic distance (CM)

Sentinel	Proxy	Chr.	Proxy Pos.	Distance (bp)	LD r²	LD D	LD D'	Proxy Allele B	Allele B Frequency
rs1478782	rs11684677	2	85,083,997	-20,150	0.97	0.11	1	C	0.126
rs1478782	rs6753370	2	85,084,192	-19,955	0.99	0.11	1	G	0.124
rs1478782	rs1009568	2	85,085,266	-18,881	0.99	0.11	1	G	0.124
rs1478782	rs11687943	2	85,085,691	-18,456	0.98	0.11	1	G	0.125
rs1478782	rs6744598	2	85,085,692	-18,455	0.99	0.11	1	A	0.124
rs1478782	rs7593934	2	85,086,038	-18,109	0.99	0.11	1	C	0.124
rs1478782	rs55906098	2	85,087,192	-16,955	0.99	0.11	1	G	0.124
rs1478782	rs76541252	2	85,088,309	-15,838	1	0.11	1	C	0.123
rs1478782	rs78839646	2	85,088,340	-15,807	1	0.11	1	C	0.123
rs1478782	rs735797	2	85,089,414	-14,733	1	0.11	1	A	0.123
rs1478782	rs75499193	2	85,089,740	-14,407	1	0.11	1	T	0.123
rs1478782	rs11683239	2	85,090,036	-14,111	0.96	0.11	1	G	0.127
rs1478782	rs12464591	2	85,091,990	-12,157	1	0.11	1	T	0.123
rs1478782	rs12465338	2	85,092,099	-12,048	1	0.11	1	G	0.123
rs1478782	rs112524336	2	85,094,533	-9,614	1	0.11	1	C	0.123
rs1478782	rs9751440	2	85,095,065	-9,082	1	0.11	1	G	0.123
rs1478782	rs1000919	2	85,096,481	-7,666	1	0.11	1	T	0.123
rs1478782	rs80090251	2	85,098,270	-5,877	1	0.11	1	C	0.123
rs1478782	rs74417790	2	85,099,029	-5,118	1	0.11	1	T	0.123
rs1478782	rs111583032	2	85,099,404	-4,743	1	0.11	1	G	0.123
rs1478782	rs1478782	2	85,104,147	0	1	1	1	A	0.123
rs1478782	rs141756221	2	85,108,309	4,162	0.99	0.11	1	C	0.124
rs1478782	rs75893711	2	85,110,397	6,250	0.99	0.11	1	C	0.124

Showing 1 to 23 of 23 entries

```
knitr::include_graphics("Screenshot 2025-05-20 002144.png")
```




```r
knitr::include_graphics("Screenshot 2025-05-20 021522.png")
```

SNP properties – Genome Assembly: grch37, Variant set: 1kgpp3v5, Population: EUR			
rs111583032			
position / outlink		**allele info**	
physical position	chr2: 85,099,404	**alleles**	A/G
genetic position [cM]	114.48	**frequencies**	0.877/0.123
outlink	e!	**VEP effect allele**	A

Basic features

Conservation/deleteriousness		**Linked genes**	
phyloP	-4.531	**gene(s) hit or close-by**	RPL12P18 e!, TRABD2A e!
phastCons	0.037	**eQTL gene(s)**	AC105053.3 e!, DNAH6 e!, SUCLG1 e!, TMSB10 e!, TRABD2A e!
GERP++	-0.47	**pQTL gene(s)**	–
CADD score	0.358	**potentially regulated gene(s)**	–
SnpEff effect impact	modifier	**disease gene(s)**	SUCLG1 e!

Trait annotations

Show 10 entries Search:

mQTL association

metabolite	biofluid	p-value	source	source	entry/link
etiocholanolone glucuronide	urine	2.80×10^{-5}	Schlosser et al.	31959995	

Showing 1 to 1 of 1 entries

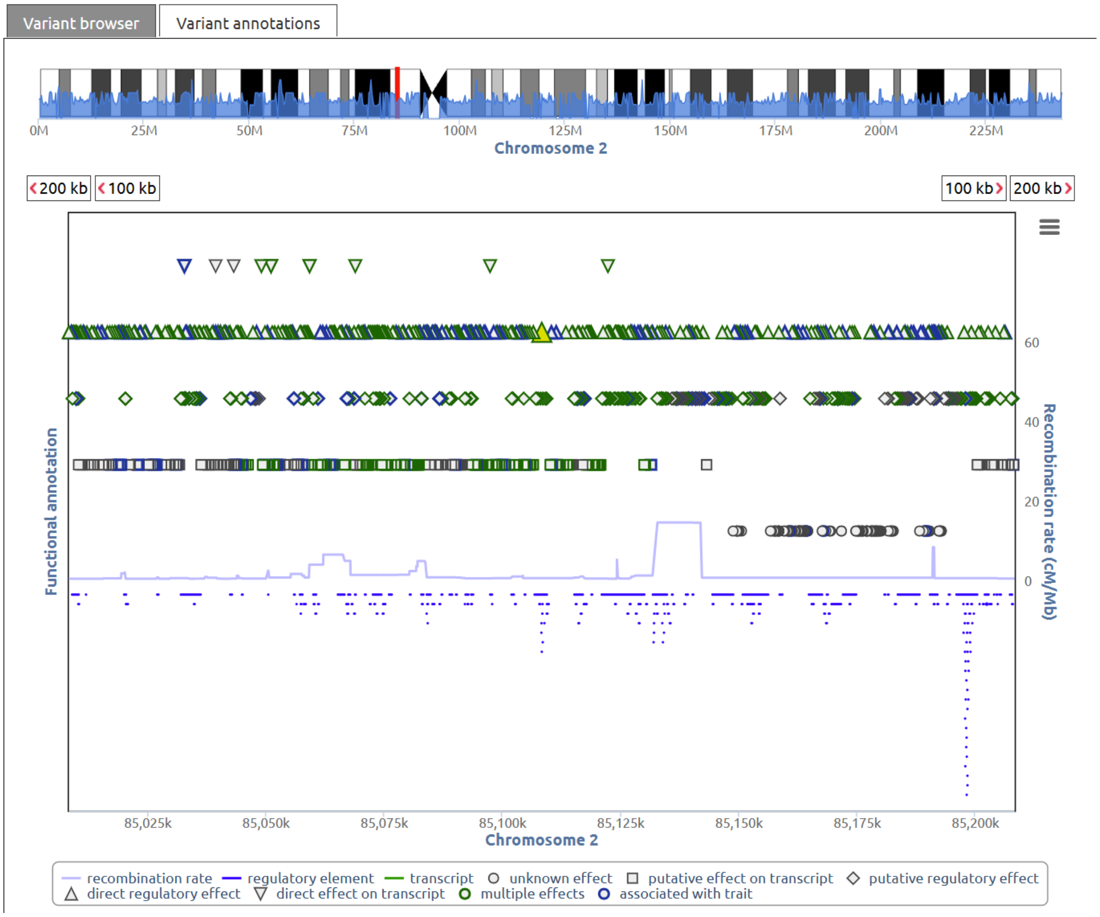
Show 10 entries Search:

Disease gene annotation

gene	trait	source DB	source	entry/link
SUCLG1 e!	MITOCHONDRIAL DNA DEPLETION SYNDROME 9 ENCEPHALOMYOPATHIC TYPE [...]	OMIM	MIM:245400	OMIM
SUCLG1 e!	Fatal Infantile Lactic Acidosis	DECIPHER	MIM:308078	OMIM
SUCLG1 e!	Fatal infantile lactic acidosis with methylmalonic aciduria	OrphaNet	OrphaNet:17	orphanet

Showing 1 to 3 of 3 entries

```r
knitr::include_graphics("Screenshot 2025-05-20 002303.png")
```




```r
knitr::include_graphics("Screenshot 2025-05-20 021611.png")
```

SNP properties – Genome Assembly: grch37, Variant set: 1kgpp3v5, Population: EUR

rs141756221

position / outlink		allele info	
physical position	chr2: 85,108,309	**alleles**	T/C
genetic position [cM]	114.48	**frequencies**	0.876/0.124
outlink	e!	**VEP effect allele**	T

Basic features

Conservation/deleteriousness		Linked genes	
phyloP	-1.062	**gene(s) hit or close-by**	TRABD2A e!
phastCons	0	**eQTL gene(s)**	DNAH6 e!, SUCLG1 e!, TMSB10 e!, TRABD2A e!
GERP++	0.552	**pQTL gene(s)**	–
CADD score	7.947	**potentially regulated gene(s)**	TRABD2A e!
SnpEff effect impact	modifier	**disease gene(s)**	SUCLG1 e!

Trait annotations

Show 10 entries

Search:

Disease gene annotation

gene	trait	source DB	source	entry/link
SUCLG1 e!	MITOCHONDRIAL DNA DEPLETION SYNDROME 9 ENCEPHALOMYOPATHIC TYPE [...]	↗ OMIM	MIM:245400	OMIM®
SUCLG1 e!	Fatal Infantile Lactic Acidosis	↗ DECIPHER	MIM:308078	OMIM®
SUCLG1 e!	Fatal infantile lactic acidosis with methylmalonic aciduria	↗ OrphaNet	OrphaNet:17	orphanet

Showing 1 to 3 of 3 entries